

Chemistry — class test

Chemistry (4CH1) · IGCSE · total 304 marks · name: _____

Question 1 (1 mark)

(a)

[1]

Propanoic acid reacts with ethanol in an esterification reaction.

What is the displayed formula of the ester that is formed during this reaction?

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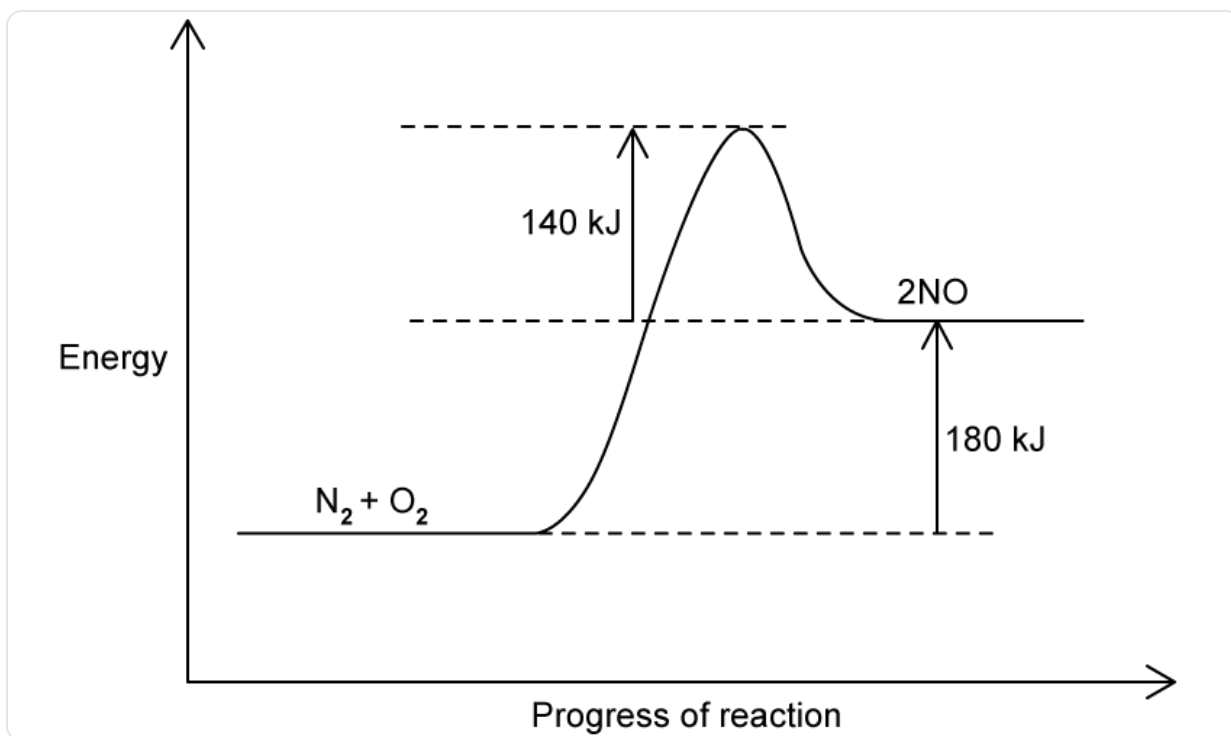
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Question 2 (1 mark)

(a)

[1]

The reaction profile for making nitrogen monoxide, NO, is shown below:



What is the activation energy for this reaction?

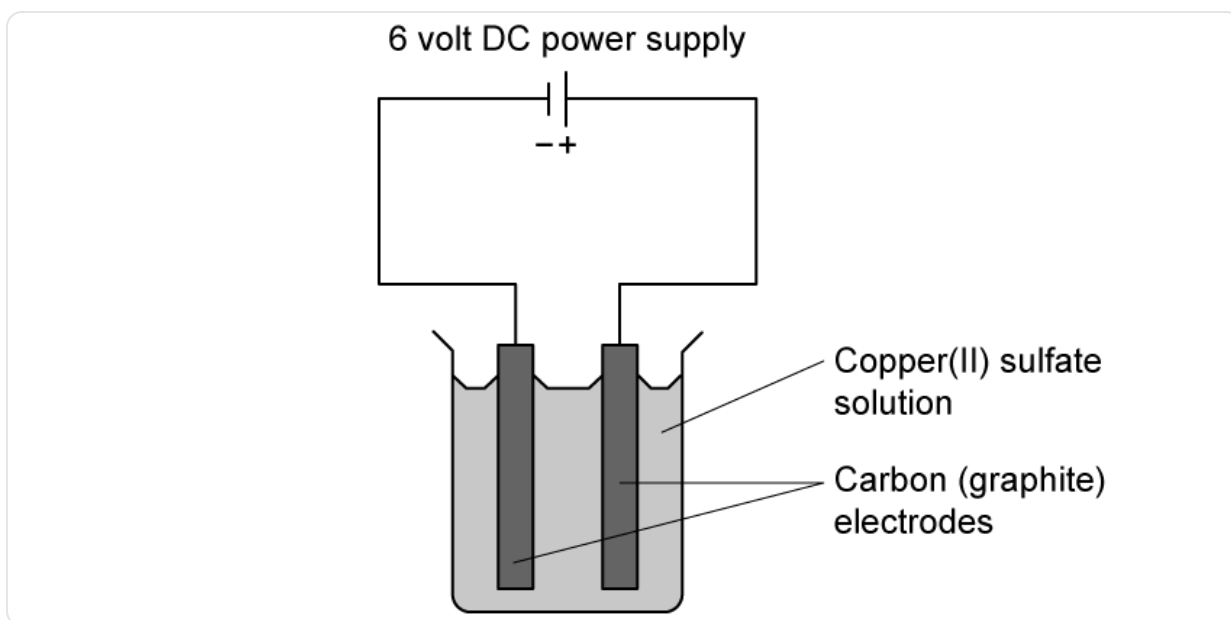
- A) +140 kJ
- B) +180 kJ
- C) +320 kJ
- D) -330 kJ

Question 3 (1 mark)

(a)

[1]

A student uses the apparatus to electrolyse an aqueous solution of copper(II) sulfate.



Which substances are formed at the negative electrode (cathode) and the positive electrode (anode)?

- A) Negative electrode: hydrogen Positive electrode: sulfur
- B) Negative electrode: copper Positive electrode: oxygen
- C) Negative electrode: sulfur Positive electrode: hydrogen
- D) Negative electrode: oxygen Positive electrode: copper

Question 4 (1 mark)

(a)

[1]

A company produces tin from an ore called cassiterite, which is mainly tin oxide.

What is the relative formula mass (M_r) of tin oxide, SnO_2 ?

[A_r of Sn = 119; O = 16]

- A) 135
- B) 151
- C) 270
- D) 1904

Question 5 (1 mark)

(a)

[1]

A student is carrying out a titration using phenolphthalein indicator.

What is the correct colour of this indicator in an acid and alkali?

- A) Colour in acid: pink Colour in alkali: colourless
- B) Colour in acid: colourless Colour in alkali: pink
- C) Colour in acid: red Colour in alkali: yellow
- D) Colour in acid: orange Colour in alkali: red

Question 6 (1 mark)

(a)

[1]

Which element in Group 7 is a solid at room temperature?

- A) fluorine
- B) chlorine
- C) bromine
- D) iodine

Question 7 (1 mark)

(a)

[1]

A compound is suspected of containing copper ions, Cu^{2+} and bromide ions, Br^- .

Which test would **not** help to positively identify copper ions or bromide ions?

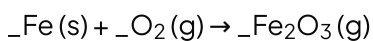
- A) Test: flame test Result: blue-green flame
- B) Test: add sodium hydroxide solution Result: blue precipitate forms
- C) Test: add acidified silver nitrate solution Result: cream precipitate forms
- D) Test: add sodium hydroxide solution Result: green precipitate forms

Question 8 (1 mark)

(a)

[1]

What numbers will balance the following equation?



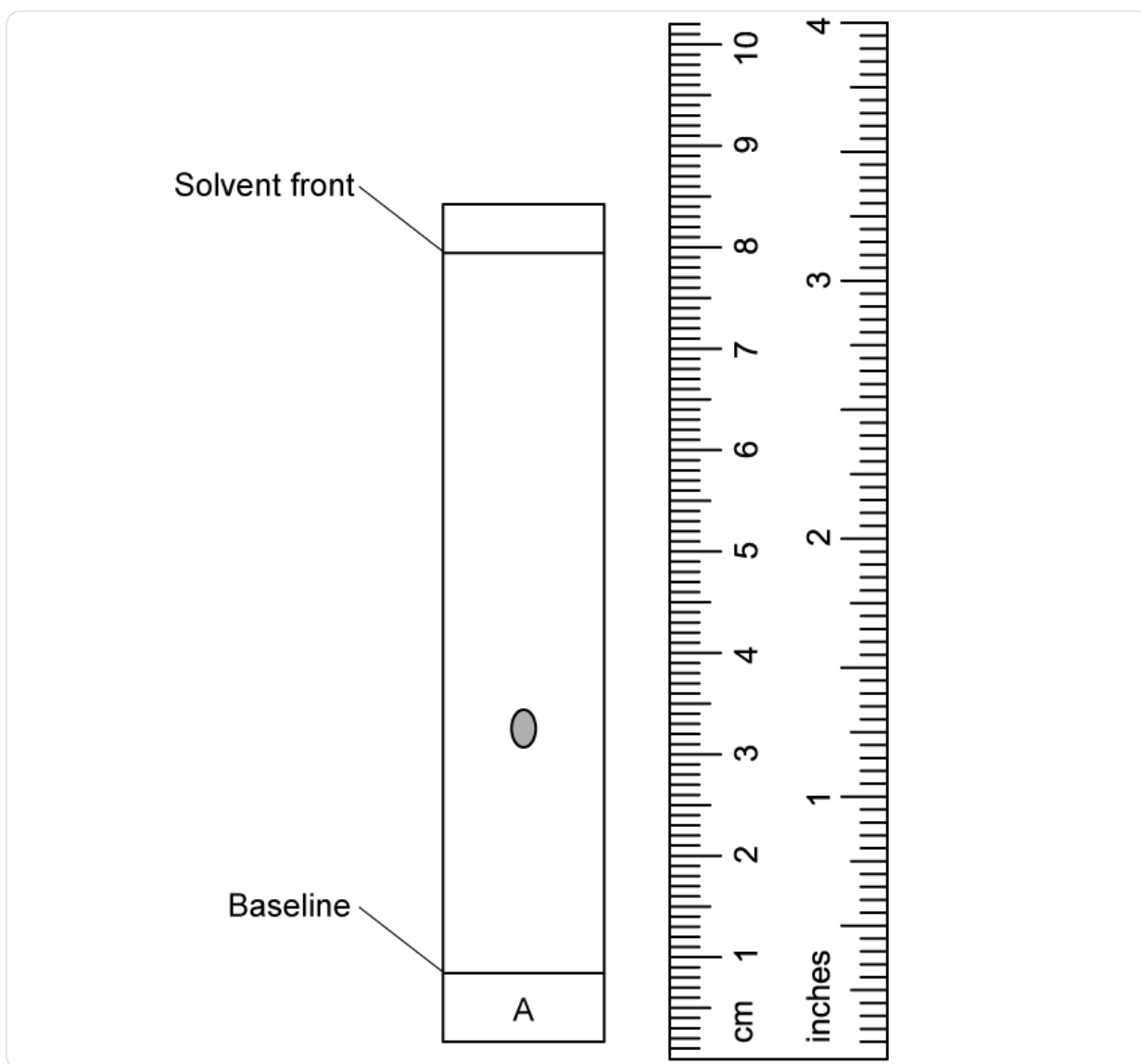
- A) 2,3,4
- B) 4,3,2
- C) 2,3,1
- D) 1,2,3

Question 9 (1 mark)

(a)

[1]

A student obtained the following chromatogram for substance A.



Determine the R_f value of substance A.

- A) 0.66
- B) 0.26
- C) 0.35
- D) 0.43

Question 10 (1 mark)

(a)

[1]

What type of reaction occurs when iron rusts?

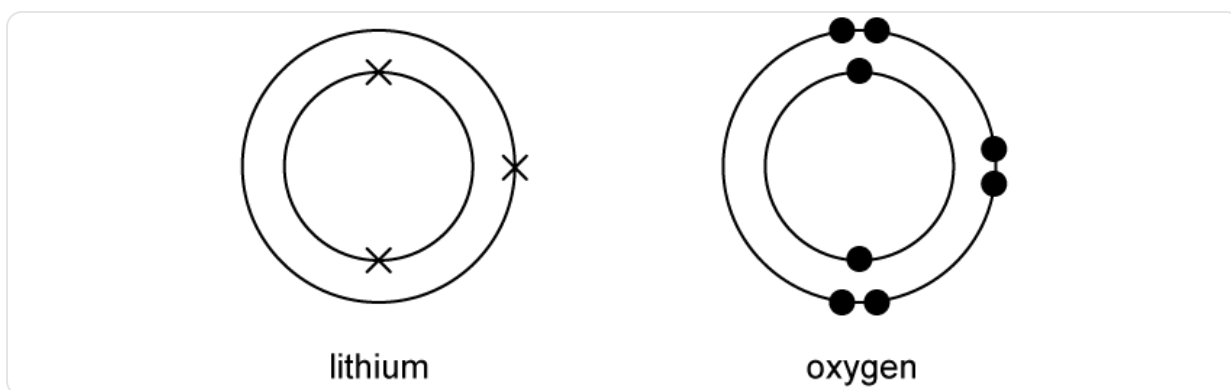
- A) Oxidation
- B) Neutralisation
- C) Combustion
- D) Displacement

Question 11 (1 mark)

(a)

[1]

The diagram shows the electronic configurations of an atom of lithium and an atom of oxygen.



Which statement is **not** correct about the changes in electronic configuration that occur when lithium and oxygen react to form lithium oxide, Li_2O ?

- A) Two lithium atoms each lose one electron
- B) One oxygen atom gains 2 electrons
- C) A lithium ion has an electronic configuration of 2
- D) An oxide ion has an electronic configuration of 2.6

Question 12 (1 mark)

(a)

[1]

Carbonic acid is found in fizzy drinks and has the formula, H_2CO_3 .

Which ion causes the drink to be acidic?

- A) OH^-
- B) CO_3^{2-}
- C) H^+
- D) O^{2-}

Question 13 (1 mark)

(a)

[1]

Carbon dioxide contains covalent bonds.

The covalent bonds in a molecule of carbon dioxide can be represented by a dot-and-cross diagram.

Which is the correct dot-and-cross diagram for carbon dioxide?

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Question 14 (1 mark)

(a)

[1]

Which of the following reactions is an example of an addition reaction?

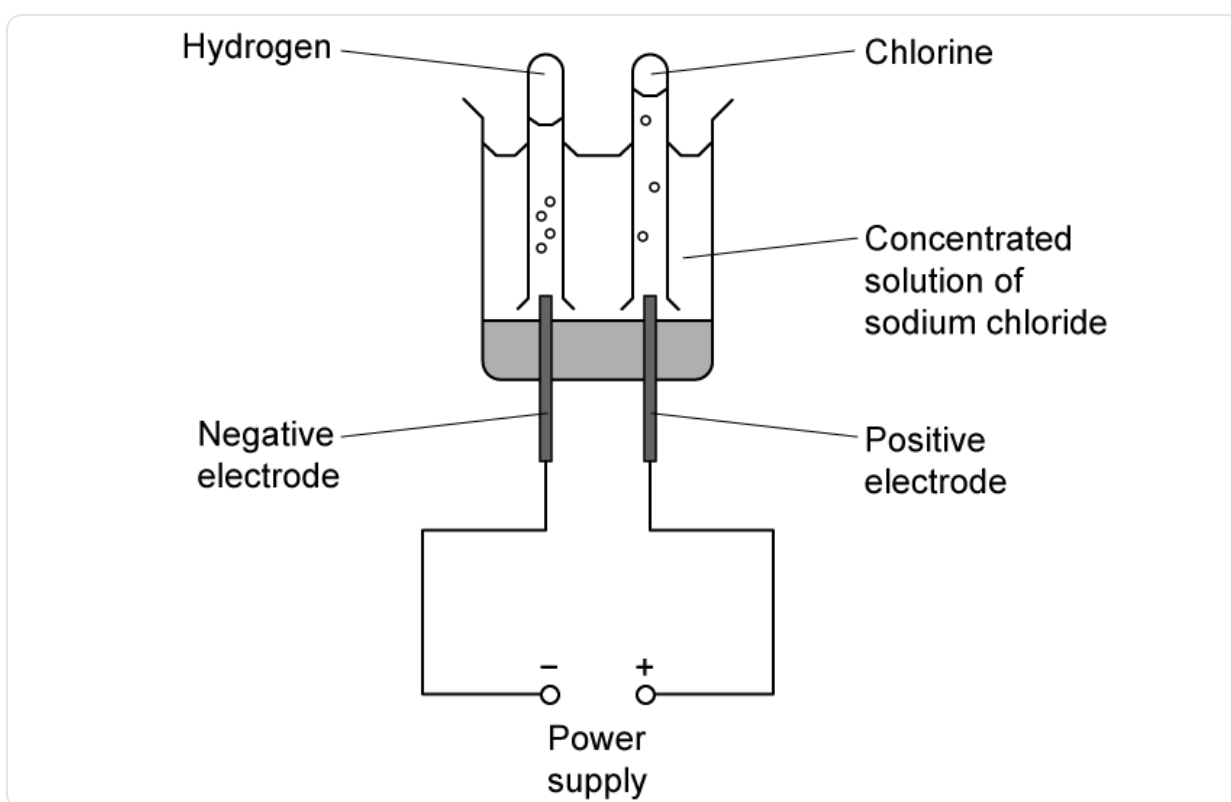
- A) $\text{CH}_4 + \frac{1}{2}\text{O}_2 \rightarrow \text{CO} + 2\text{H}_2\text{O}$
- B) $\text{C}_2\text{H}_4 + \text{Br}_2 \rightarrow \text{C}_2\text{H}_4\text{Br}_2$
- C) $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
- D) $\text{CH}_4 + \text{Br}_2 \rightarrow \text{CH}_3\text{Br} + \text{HBr}$

Question 15 (1 mark)

(a)

[1]

Hydrogen gas and chlorine gas can be prepared in a laboratory by electrolysis of a concentrated solution of sodium chloride.



Which is the ionic half-equation that represents the formation of chlorine gas at the positive electrode?

- A) $\text{Cl}^- \rightarrow \text{Cl} + \text{e}^-$
- B) $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$
- C) $\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$
- D) $2\text{Cl}^- + 2\text{e}^- \rightarrow \text{Cl}_2$

Question 16 (1 mark)

(a)

[1]

Electrolysis can be used to separate molten lead bromide.

Which statement is true for what happens at the negative electrode?

☐	A	Lead atoms are oxidised
☐	B	Lead atoms are reduced
☐	C	Lead ions are oxidised
☐	D	Lead ions are reduced

- A) Lead atoms are oxidized
- B) Lead atoms are reduced
- C) Lead ions are oxidized
- D) Lead ions are reduced

Question 17 (1 mark)

(a)

[1]

Which of the following reactions does **not** involve the oxidation of ethanol?

- A) Production of ethanol from the fermentation of glucose
- B) Heating ethanol with potassium dichromate(VI) in dilute sulfuric acid
- C) Reaction of ethanol with oxygen in the air
- D) Formation of carbon dioxide and water from the complete combustion of ethanol

Question 18 (1 mark)

(a)

[1]

Which equation does **not** show the correct reaction of an acid?

- A) copper oxide + hydrochloric acid → copper chloride + water
- B) calcium carbonate + nitric acid → calcium nitrate + carbon dioxide
- C) potassium hydroxide + sulfuric acid → potassium sulfate + water
- D) zinc + sulfuric acid → zinc sulfate + hydrogen

Question 19 (1 mark)

(a)

[1]

Fractional distillation is the process by which crude oil is separated into simpler, more useful fractions.

Which of the following statements about the process of fractional distillation is correct?

- A) The fractionating column is cooler at the bottom than at the top
- B) The fractions collected at the top of the fractionating column have a high boiling point
- C) The molecules in each fraction have a similar boiling point and carbon chain length
- D) The fractions collected at the bottom of the fractionating column have a low viscosity

Question 20 (1 mark)

(a)

[1]

Ethanoic acid is a member of the homologous series of carboxylic acids.

Which is the functional group of carboxylic acids?

.....

.....

.....

Question 21 (1 mark)

(a)

[1]

Which of the statements describes ionic bonding?

- A) Electrostatic attraction between atoms
- B) Electrostatic attraction between positively charged particles and delocalized electrons
- C) Electrostatic attraction between oppositely charged ions
- D) Electrostatic attraction between the nuclei of two atoms and a shared pair of electrons

Question 22 (1 mark)

(a)

[1]

A student added some sulfuric acid gradually to a solution of ammonia until it was in excess.

What happens to the pH during this reaction?

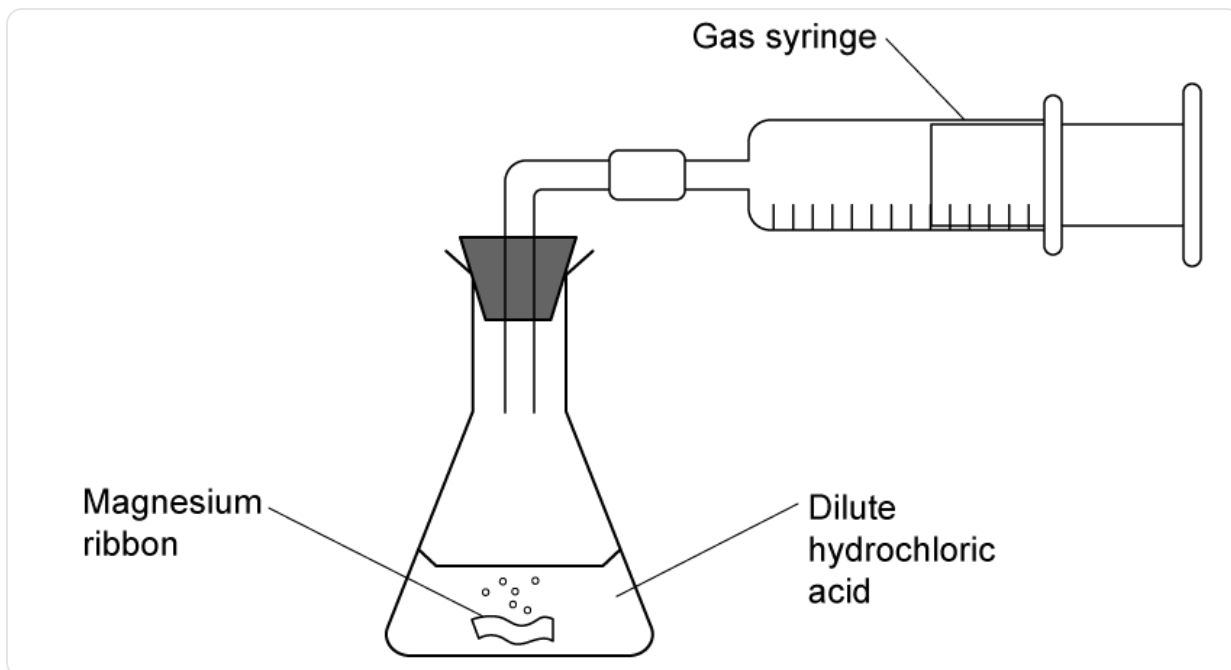
- A) pH of ammonia at start: 11 pH after excess sulfuric acid: 7
- B) pH of ammonia at start: 7 pH after excess sulfuric acid: 13
- C) pH of ammonia at start: 11 pH after excess sulfuric acid: 2
- D) pH of ammonia at start: 3 pH after excess sulfuric acid: 8

Question 23 (1 mark)

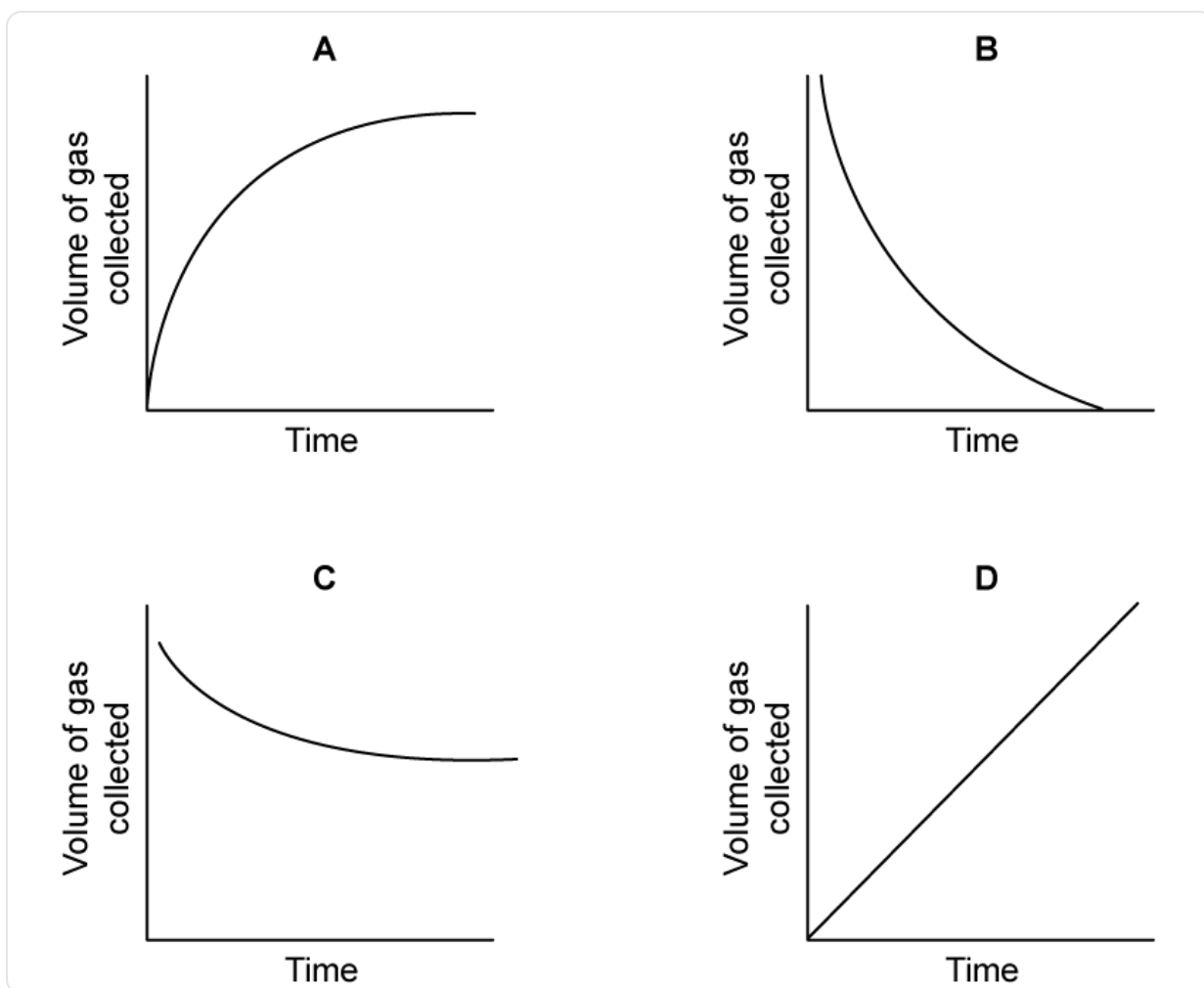
(a)

[1]

A student investigates the rate of reaction between an **excess** of hydrochloric acid and a limiting mass of magnesium using the apparatus shown:



Which of the graphs could represent her results?



Question 24 (1 mark)

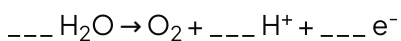
(a)

[1]

When copper(II) sulfate undergoes electrolysis, it produces oxygen gas at the positive electrode.

The unbalanced half equation below represents the reaction that occurs at the positive electrode.

Which numbers will correctly balance the half equation?



☐	A	2,2,2
☐	B	2,4,2
☐	C	2,4,4
☐	D	4,4,2

A) 2,2,2

B) 2,4,2

C) 2,4,4

D) 4,4,2

Question 25 (1 mark)

(a)

[1]

Sodium oxide reacts with water to form sodium hydroxide.

A scientist plans to measure the temperature change of the reaction using calorimetry in a lab.

This is the scientist's basic method:

- use 50 g of water
- record the temperature of the water
- add sodium hydroxide
- stir
- record the maximum temperature reached

Which apparatus list is the most appropriate for this experiment?

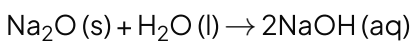
- A) Electronic balance, measuring cylinder, glass beaker, thermometer
- B) Measuring cylinder, glass beaker, polystyrene cup, thermometer
- C) Electronic balance, glass beaker, polystyrene cup, thermometer
- D) Measuring cylinder, glass beaker, glass rod, thermometer

Question 26 (1 mark)

(a)

[1]

Another student carries out the experiment to find the heat energy released in the reaction of sodium oxide with water:



They found that 10,800 J of heat energy was released when 0.060 mol of sodium oxide reacts.

What is the value of ΔH , in kJ/mol, for the reaction of sodium oxide with water?

- A) 0.0056 kJ/mol
- B) 180 kJ/mol
- C) 1,800 kJ/mol
- D) 8,000 kJ/mol

Question 27 (1 mark)

(a)

[1]

The diagram shows part of the periodic table, with elements represented by the letters M, Q, R and T. The letters in the diagram represent elements but are not their chemical symbols.

Use the Periodic Table to help you answer this question.

																M	
	Q											R					
T																	

Which of these statements is true?

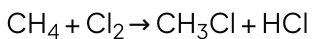
- A) T and Q form acidic oxides
- B) M conducts electricity
- C) M is a non-metal element
- D) R is a gaseous non-metal element

Question 28 (1 mark)

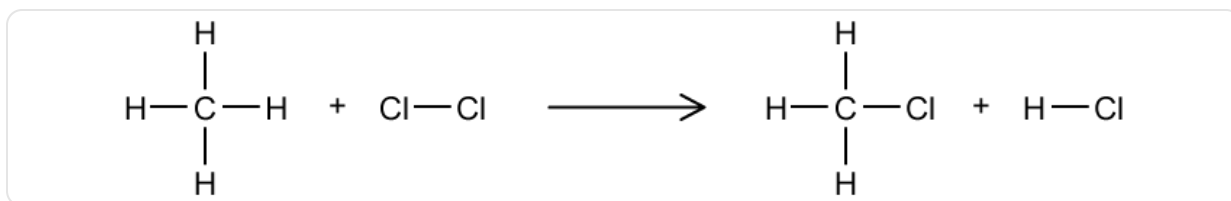
(a)

[1]

Chlorine reacts with methane in sunlight.



The diagram below shows the displayed formulae for the reaction of chlorine with methane.



The table below shows the bond energies and the overall energy change in the reaction.

	C-H	Cl-Cl	C-Cl	H-Cl	Overall energy change
Energy in kJ/mol	413	243	X	432	-122

What is the value of X?

- A) 67
- B) 134
- C) 346
- D) 542

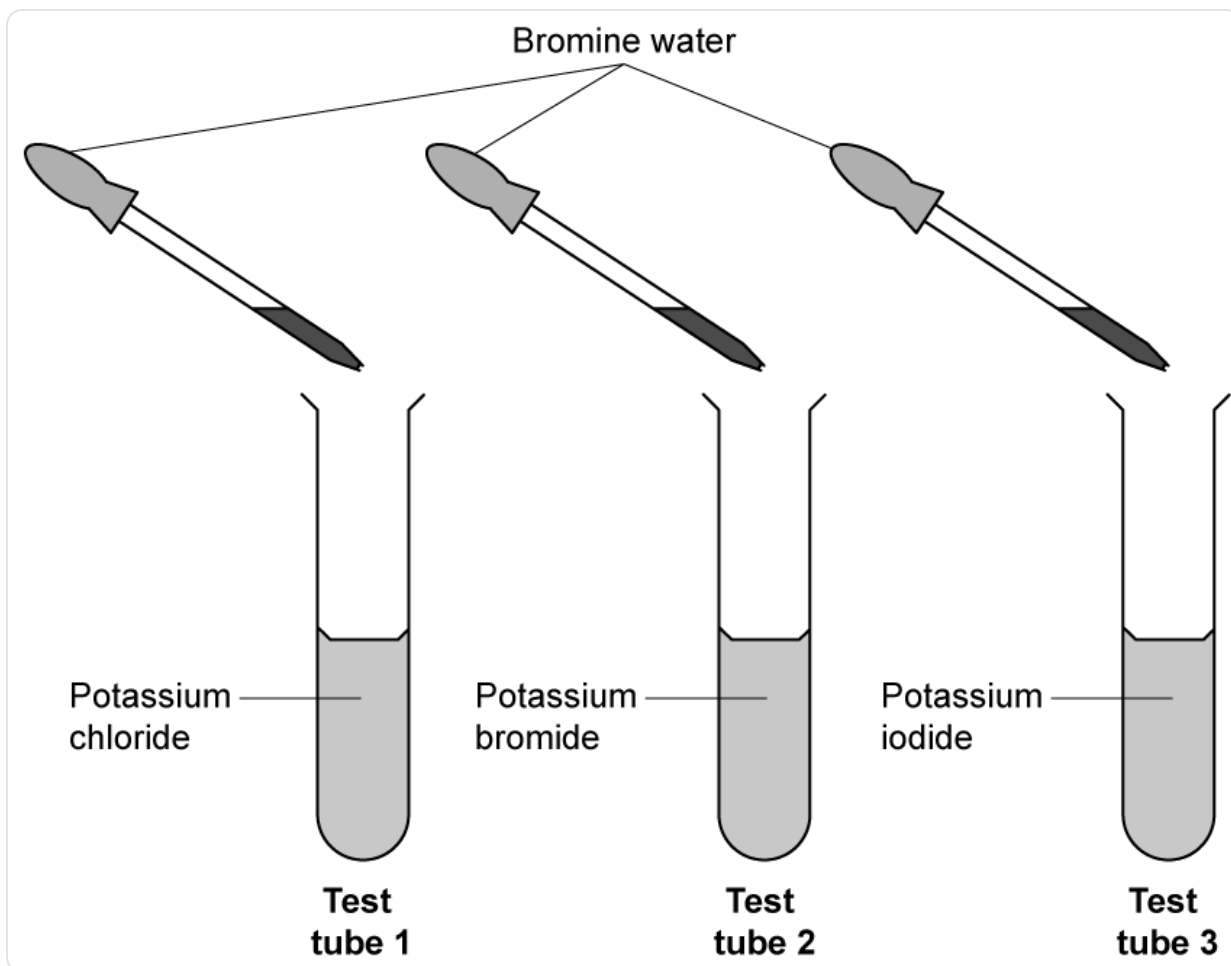
Question 29 (1 mark)

(a)

[1]

Bromine water is an aqueous solution of bromine which is orange in colour.

What would be seen if bromine water was added to the test tubes in the diagram?



	Test tube 1	Test tube 2	Test tube 3
☐ A	solution stays orange	solution turns brown	solution turns brown
☐ B	solution turns pale green	solution stays orange	solution turns colourless
☐ C	solution stays orange	solution stays orange	solution turns brown
☐ D	solution turns colourless	solution turns colourless	solution turns brown

Question 30 (1 mark)

(a)

[1]

Scientists test a solution of compound X. The table shows their results.

Test	Result
Acidified silver nitrate solution	White precipitate
Add sodium hydroxide solution	Green precipitate

Which **two** ions are present in compound X?

☐	A	Fe^{3+} and Cl^-
☐	B	Fe^{3+} and SO_4^{2-}
☐	C	Fe^{2+} and Cl^-
☐	D	Fe^{2+} and SO_4^{2-}

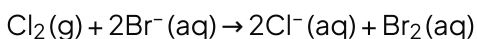
- A) Fe^{3+} and Cl^-
B) Fe^{3+} and SO_4^{2-}
C) Fe^{2+} and Cl^-
D) Fe^{2+} and SO_4^{2-}

Question 31 (1 mark)

(a)

[1]

The ionic equation for the reaction between chlorine and sodium bromide is:



This reaction occurs because chlorine is more reactive than bromine.

Which of the statements explaining why chlorine is more reactive than bromine is **not** true?

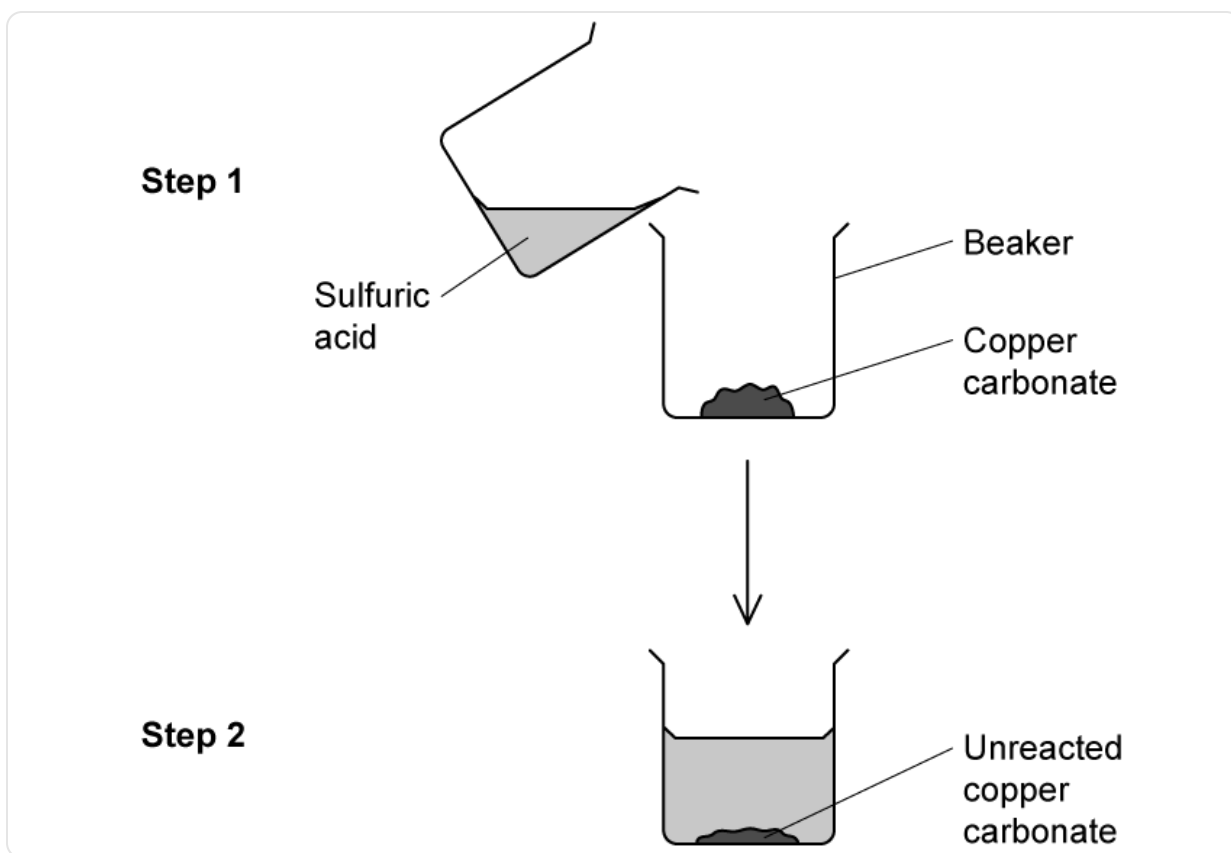
- A) chlorine has smaller atoms than bromine
B) chlorine atoms gain electrons more easily
C) chlorine atoms lose electrons more easily
D) chlorine atoms have fewer electron shells than bromine atoms

Question 32 (1 mark)

(a)

[1]

A student making copper sulfate crystals used the method below.



Unreacted copper carbonate was left over as it had been added in excess.

What is the reason for adding it in excess and what would step 3 be of this method?

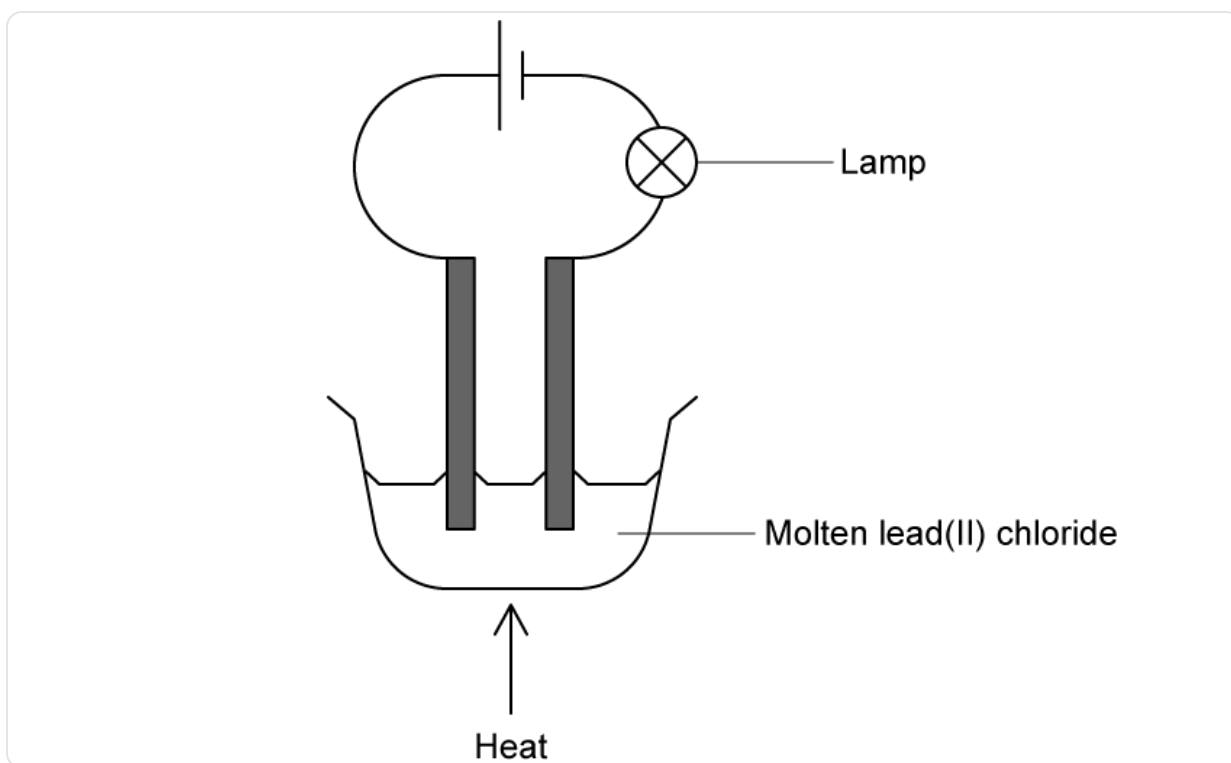
- A) Reason: to produce a greater amount of salt crystals Step 3: filtration
- B) Reason: to improve the colour intensity of the crystals Step 3: crystallisation
- C) Reason: to ensure all the acid reacts Step 3: filtration
- D) Reason: to increase the rate of reaction Step 3: evaporation

Question 33 (1 mark)

(a)

[1]

A pure sample of molten lead(II) chloride was electrolysed.



Which is the correct statement about this electrolysis?

- A) The lamp would stop working soon after the heat was removed
- B) A silvery liquid would be seen at the positive electrode
- C) Chlorine is formed at the negative electrode
- D) Hydrogen gas is formed at the negative electrode

Question 34 (1 mark)

(a)

[1]

The actual molar enthalpy change, ΔH , for the reaction of sodium oxide with water, is -238 kJ/mol .

Experimental results from calorimetry in labs give less exothermic results.

Which of the following does not explain this difference?

- A) The reaction may have been incomplete
- B) Heat energy also transfers to the surroundings
- C) The calorimetry equipment also has a specific heat capacity, and so affects the final temperature reading
- D) The student measured the initial temperature of the water as lower than the actual value.

Question 35 (1 mark)

(a)

[1]

Butanol is a member of the homologous series of alcohols.

Which of the following is **not** a formula of butanol?

- A) $C_4H_{10}O$
- B) $C_4H_{10}OH$
- C) C_4H_9OH
- D) $CH_3CH_2CH_2CH_2OH$

Question 36 (1 mark)

(a)

[1]

A neutralisation reaction occurs between ammonia and sulfuric acid.

How does the ammonia act in this reaction?

- A) As a proton donor
- B) As an electron acceptor
- C) As a proton acceptor
- D) As a neutron donor

Question 37 (1 mark)

(a)

[1]

Which statement describes metallic bonding?

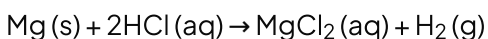
- A) Electrostatic attraction between atoms
- B) Electrostatic attraction between positively charged particles and delocalised electrons
- C) Electrostatic attraction between oppositely charged ions
- D) Electrostatic attraction between the nuclei of two atoms and a shared pair of electrons

Question 38 (1 mark)

(a)

[1]

Magnesium metal reacts with an excess of hydrochloric acid solution to form magnesium chloride and hydrogen:



Which of the following will **not** increase the rate of this reaction?

- A) Increase the temperature of the acid
- B) Increase the surface area of the pieces of magnesium
- C) Increase the concentration of the hydrochloric acid
- D) Increase the volume of hydrochloric acid solution used

Question 39 (1 mark)

(a)

[1]

Which of these statements about catalysts is **not** correct?

- A) The mass of the catalyst at the start of the reaction is the same at the end of the reaction
- B) Catalysts increase the rate of reaction but do not increase the yield
- C) Catalysts are not chemically changed during the course of the reaction
- D) Catalysts provide an alternative reaction pathway which has a lower activation energy

Question 40 (1 mark)

(a)

[1]

What is the relative formula mass of hydrated magnesium carbonate, $\text{MgCO}_3 \cdot 2\text{H}_2\text{O}$?

[A_r of Mg = 24; H = 1; O = 16; C = 12]

- A) 84
- B) 102
- C) 112
- D) 120

Question 41 (1 mark)

(a)

[1]

Methane and bromine react to form bromomethane and hydrogen bromide.

What does this reaction occur in the presence of?

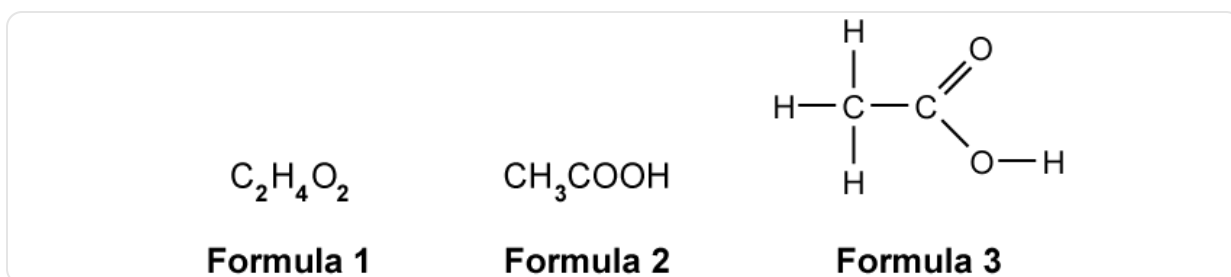
- A) Infrared radiation
- B) Visible light
- C) Ultraviolet radiation
- D) Ozone

Question 42 (1 mark)

(a) what

[1]

The formula for ethanoic acid can be written in different ways:



What are the correct names for the different ways to represent formulae?

		Formula 1	Formula 2	Formula 3
☐	A	molecular formula	structural formula	displayed formula
☐	B	structural formula	molecular formula	displayed formula
☐	C	molecular formula	empirical formula	structural formula
☐	D	empirical formula	structural formula	molecular formula

Question 43 (1 mark)

(a)

[1]

Which statement is **incorrect** about testing a sample of suspected pure water?

- A) Anhydrous copper(II) sulfate would turn blue when the solution was added
- B) The resulting hydrated copper(II) sulfate would have the chemical formula $\text{CuSO}_4 \cdot \text{H}_2\text{O}$
- C) The boiling point should be 100°C
- D) The boiling point should be sharp, not gradual

Question 44 (1 mark)

(a)

[1]

What is the maximum volume, in dm^3 , of chlorine gas at rtp that can be obtained from 17.6 tonnes of molten potassium chloride?

[1 tonne = 10^6 g]

[M_r of KCl = 74.5]

[Assume that the molar volume of chlorine at rtp is $24 \text{ dm}^3/\text{mol}$]

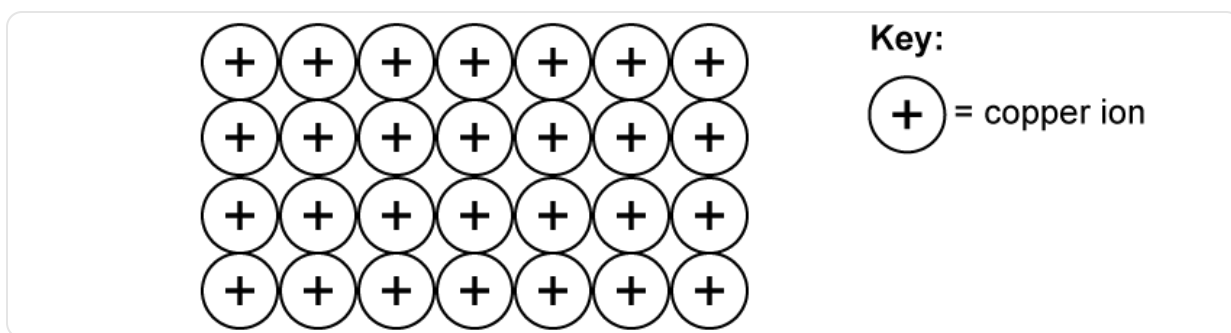
- A) 2.83
- B) 5.67
- C) 2.83×10^6
- D) 5.67×10^6

Question 45 (1 mark)

(a)

[1]

The diagram shows the arrangement of ions in copper metal.



Which statement explains a physical property of copper?

- A) Copper can conduct electricity as its ions are able to carry charge around its structure
- B) Copper has a high melting point as the metal ions are strongly attracted to each other
- C) Copper is malleable as the layers of ions can slide over each other
- D) Copper is malleable as the metallic bonds between the layers of ions break easily

Question 46 (1 mark)

(a)

[1]

How much energy is needed to break all the bonds in 0.025 g of nitrogen gas?

The relative atomic mass, A_r , of N is 14.

Bond	Bond energy in kJ/mol
$\text{N}\equiv\text{N}$	945

- A) 1701 J
- B) 1701 kJ
- C) 844 J
- D) 844 kJ

Question 47 (1 mark)

(a)

[1]

What is the pH of a weak acid?

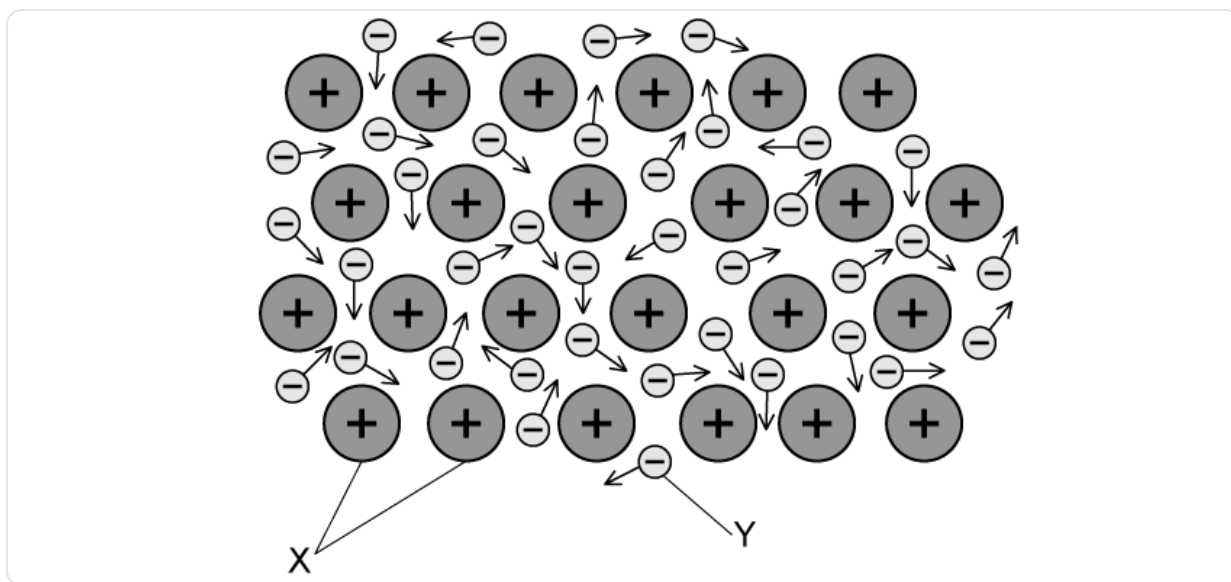
- A) 2
- B) 6
- C) 7
- D) 11

Question 48 (1 mark)

(a)

[1]

A metallic lattice structure is shown in the diagram.



What are the missing labels, X & Y?

- A) X: positively charged metal ions Y: negatively charged ions
- B) X: positively charged protons Y: negatively charged non-metal ions
- C) X: positively charged metal ions Y: delocalised electrons
- D) X: positively charged protons Y: negatively charged electrons

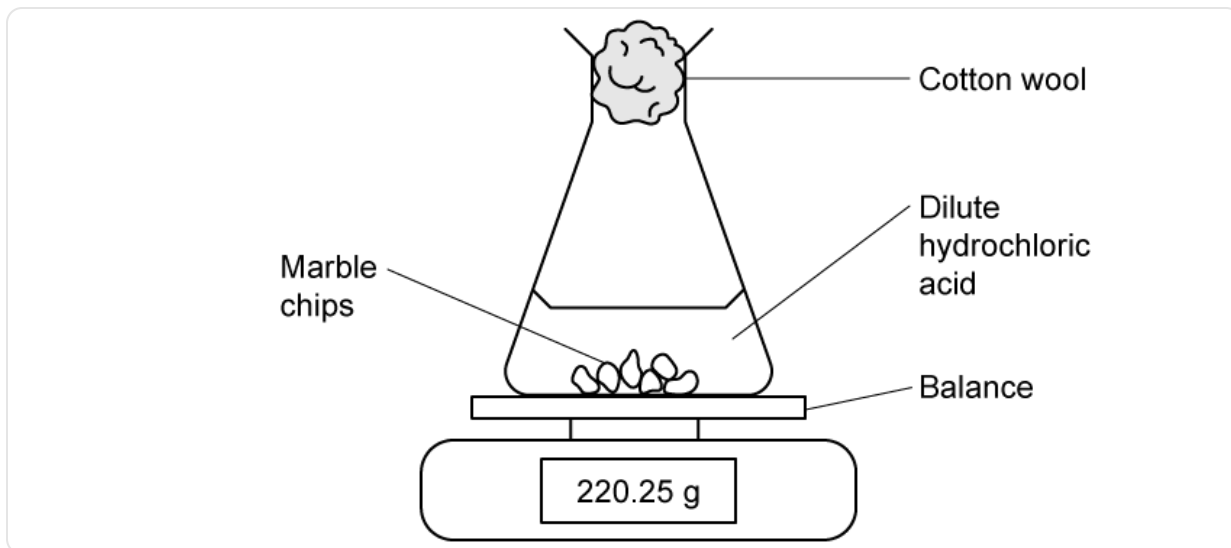
Question 49 (1 mark)

(a)

[1]

Some students investigate the rate of reaction between marble chips and hydrochloric acid.

The diagram shows the apparatus they use.



What is the purpose of the cotton wool?

- A) To prevent carbon dioxide gas from escaping
- B) To ensure the mass stays constant throughout the experiment
- C) To prevent the acid from leaving the conical flask
- D) To prevent oxygen entering the conical flask

Question 50 (1 mark)

(a)

[1]

Calcium carbonate reacts with hydrochloric acid to produce calcium chloride, water and carbon dioxide:



Which statement is **not** true if the temperature of the reaction decreases?

- A) The frequency of the collisions between particles decreases
- B) The collisions are less energetic
- C) The activation energy increases
- D) There are fewer particles with the minimum energy to react

Question 51 (1 mark)

(a)

[1]

Which is true about chemical energy changes?

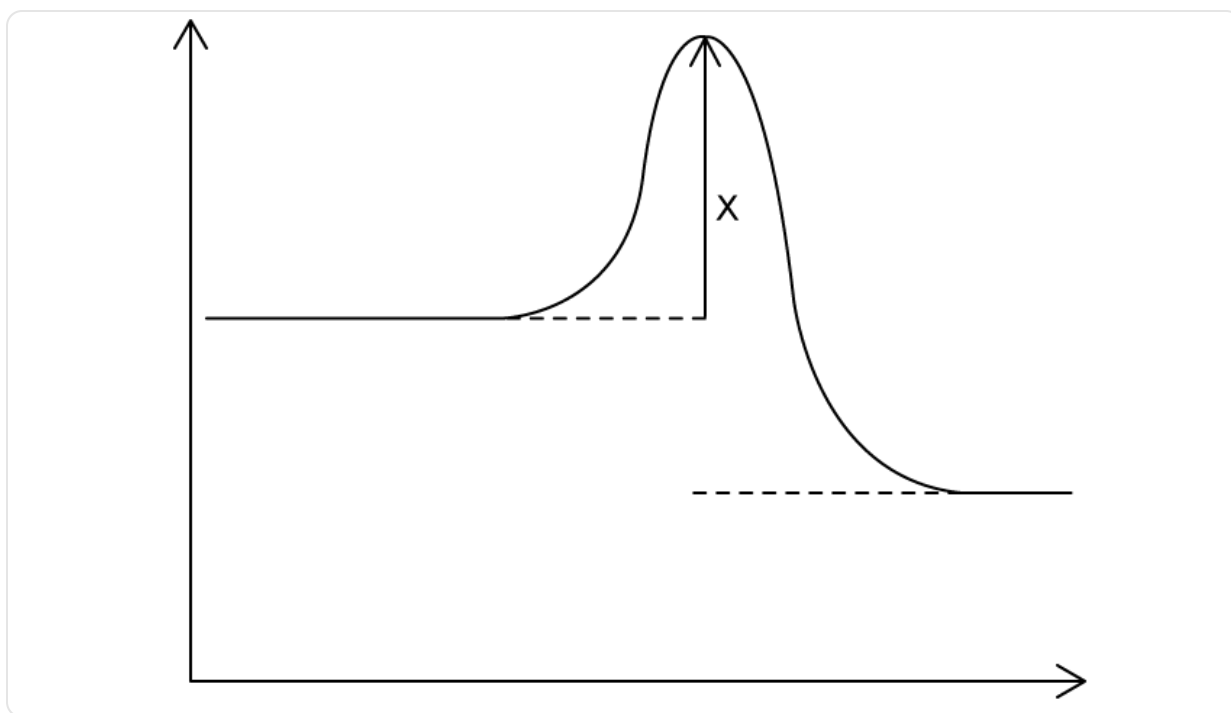
	Reaction type	Energy flows	Temperature of surroundings
A	exothermic	to the surroundings	increases
B	exothermic	from the surroundings	decreases
C	endothermic	to the surroundings	decreases
D	endothermic	from the surroundings	increases

Question 52 (1 mark)

(a)

[1]

The graph below shows a reaction profile for the reaction between methane and oxygen.



What does the label X represent?

- A) Activation Energy
- B) Overall energy change
- C) Reactants
- D) Progress of the reaction

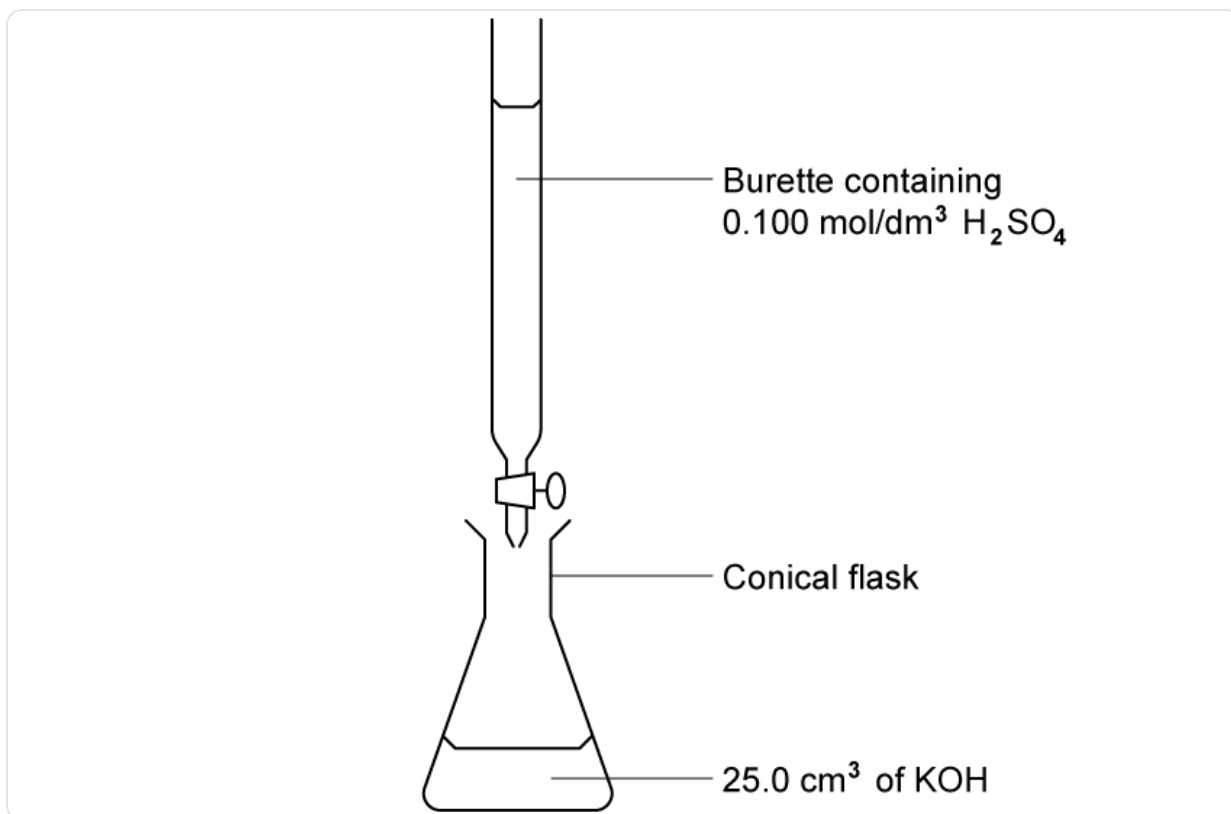
Question 53 (1 mark)

(a)

[1]

A student carried out a titration to investigate the volume of sulfuric acid needed to neutralise 25.0 cm^3 of potassium hydroxide.

The diagram shows the equipment they used.



Which piece of apparatus would the student use to measure out 25.0 cm^3 of potassium hydroxide?

- A) Pipette
- B) Measuring cylinder
- C) Beaker
- D) Conical flask

Question 54 (1 mark)

(a)

[1]

What is the formula of potassium sulfate?

- A) $\text{K}_2(\text{SO}_4)_2$
- B) $\text{K}(\text{SO}_4)_2$
- C) K_2SO_4
- D) KSO_4

Question 55 (1 mark)

(a)

[1]

Methyl orange is used as an indicator.

What is the colour of it in an alkaline solution?

- A) Yellow
- B) Orange
- C) Red
- D) Blue

Question 56 (1 mark)

(a)

[1]

The steps below briefly describe how to carry out a titration with sodium hydroxide and hydrochloric acid.

1. Add a few drops of a suitable indicator to the solution
2. Record the final volume of HCl and repeat to obtain concordant results
3. Fill the burette with HCl and record the volume
4. Measure 25 cm³ sodium hydroxide solution into a conical flask using a pipette
5. Add the HCl to sodium hydroxide until the end point is reached

What is the correct order of steps?

- A) 1, 3, 4, 5, 2
- B) 4, 1, 3, 5, 2
- C) 4, 2, 5, 1, 3
- D) 3, 1, 5, 2, 4

Question 57 (1 mark)

(a)

[1]

Which metal is most suitable for galvanizing iron?

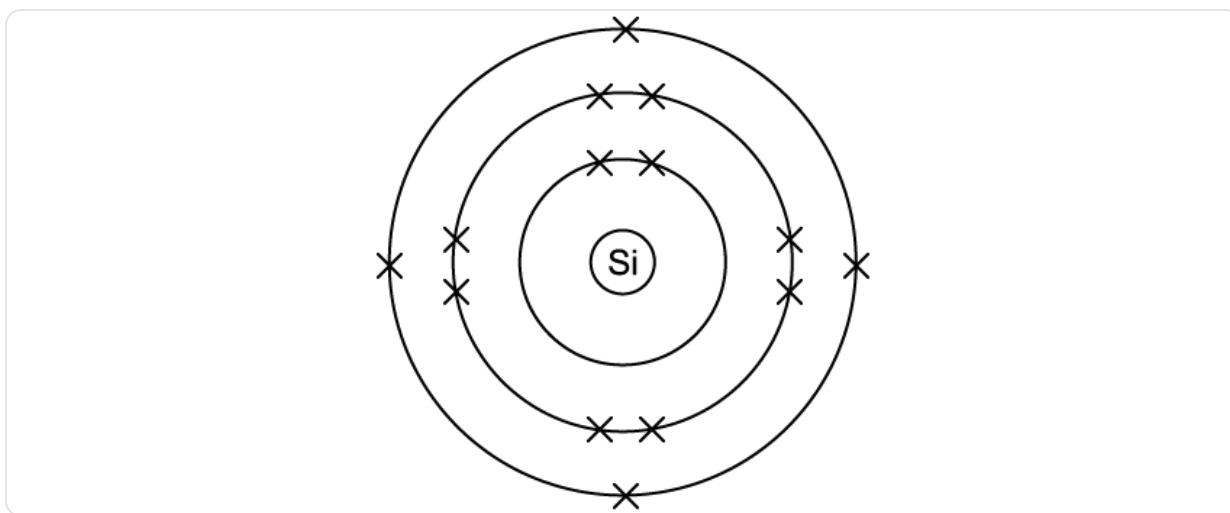
- A) Lead
- B) Copper
- C) Sodium
- D) Zinc

Question 58 (1 mark)

(a)

[1]

What is the electronic configuration of silicon?



- A) 2,4
- B) 2,2,4
- C) 2,2,8
- D) 2,8,4

Question 59 (1 mark)

(a)

[1]

Cars running on fossil fuels contribute a high amount of pollutants into the atmosphere, one of which is carbon dioxide.

Which other gas is produced by the internal combustion engine in a car?

- A) Ammonia
- B) Argon
- C) Oxides of nitrogen
- D) Nitrogen

Question 60 (1 mark)

(a)

[1]

Which of the following is **not** a feature of a homologous series?

- A) Compounds have the same functional group
- B) Compounds have the same physical properties
- C) Compounds have the same general formula
- D) Compounds have similar chemical properties

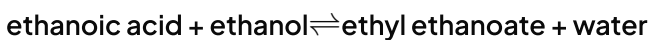
Question 61 (1 mark)

(a)

[1]

In a laboratory, ethyl ethanoate can be prepared by heating ethanol, ethanoic acid and sulfuric acid.

The following reaction occurs:



Which of the following terms cannot be used to describe the reaction that takes place?

- A) Esterification
- B) Condensation
- C) Reversible
- D) Thermal decomposition

Question 62 (1 mark)

(a)

[1]

Calculate the concentration, in mol / dm³, of a solution produced when 6.9 g of potassium carbonate, K₂CO₃, dissolves in 200 cm³ of water.

[*M_r* of K₂CO₃ = 138]

- A) 0.05 mol / dm³
- B) 0.20 mol / dm³
- C) 0.25 mol / dm³
- D) 0.50 mol / dm³

Question 63 (1 mark)

(a)

[1]

Which of these formulae is that of an alkane?

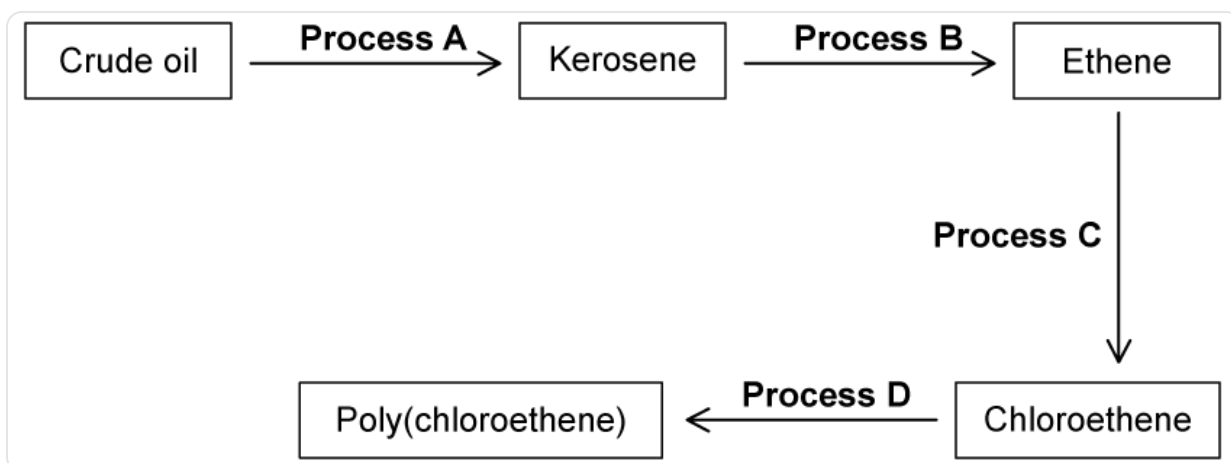
- A) C₆H₁₁
- B) C₈H₁₆
- C) C₁₀H₂₂
- D) C₁₂H₂₈

Question 64 (1 mark)

(a)

[1]

The diagram shows some processes involved in the oil industry.



Which process represents cracking?

.....

.....

.....

Question 65 (1 mark)

(a)

[1]

Ammonia, NH_3 , is a small covalent molecule.

How many **non-bonding**, outer-shell electrons are present in one molecule of ammonia?

- A) 0
- B) 2
- C) 5
- D) 6

Question 66 (1 mark)

(a)

[1]

A displacement reaction occurs between zinc and copper chloride.

Which half equation correctly shows the change that zinc undergoes?

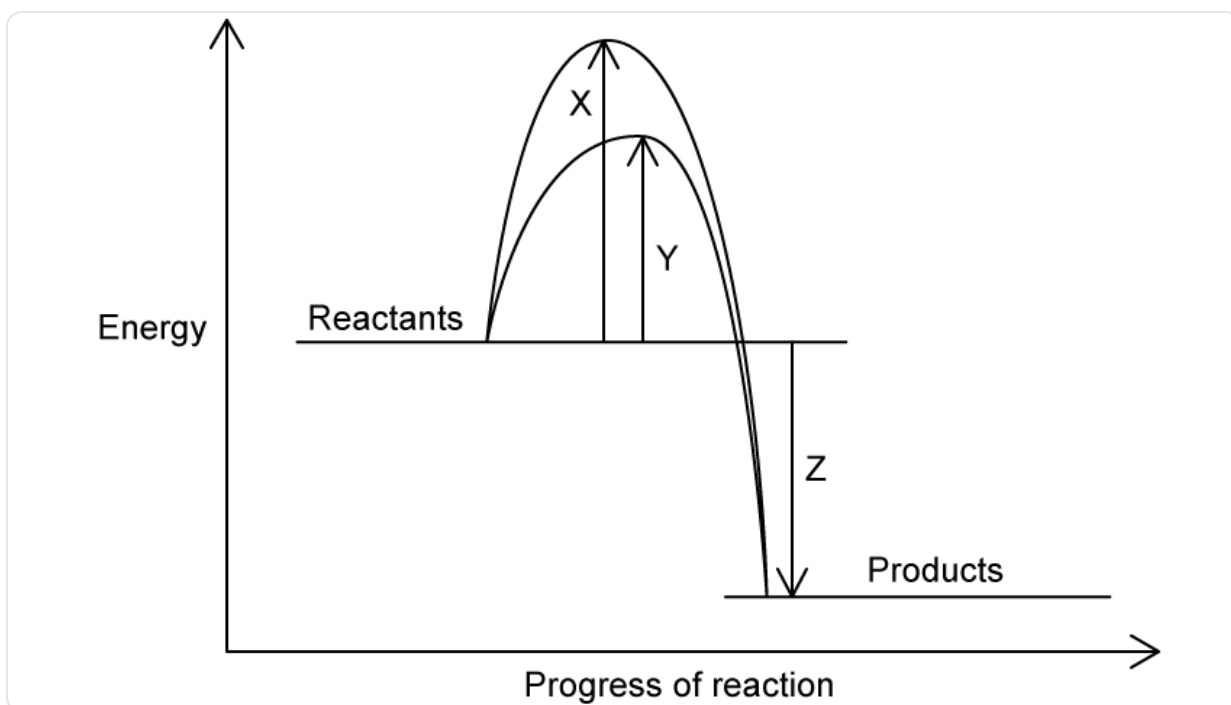
- A) $\text{Zn}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Zn}(\text{s})$
- B) $\text{Zn}(\text{s}) + 2\text{e}^{-} \rightarrow \text{Zn}^{2+}(\text{aq})$
- C) $\text{Zn}^{2+}(\text{aq}) \rightarrow \text{Zn}(\text{s}) + 2\text{e}^{-}$
- D) $\text{Zn}(\text{s}) \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^{-}$

Question 67 (1 mark)

(a)

[1]

A student draws the reaction profiles shown below for a reaction both with and without a catalyst.



Which statement is correct about the reaction profile?

- A) **X** shows the activation energy of the catalysed reaction
- B) **Z** shows ΔH
- C) The activation energy of the uncatalysed reaction pathway is less than that of the catalysed reaction pathway
- D) The reaction is endothermic

Question 68 (1 mark)

(a)

[1]

Which statement is **true** about exothermic and endothermic reactions?

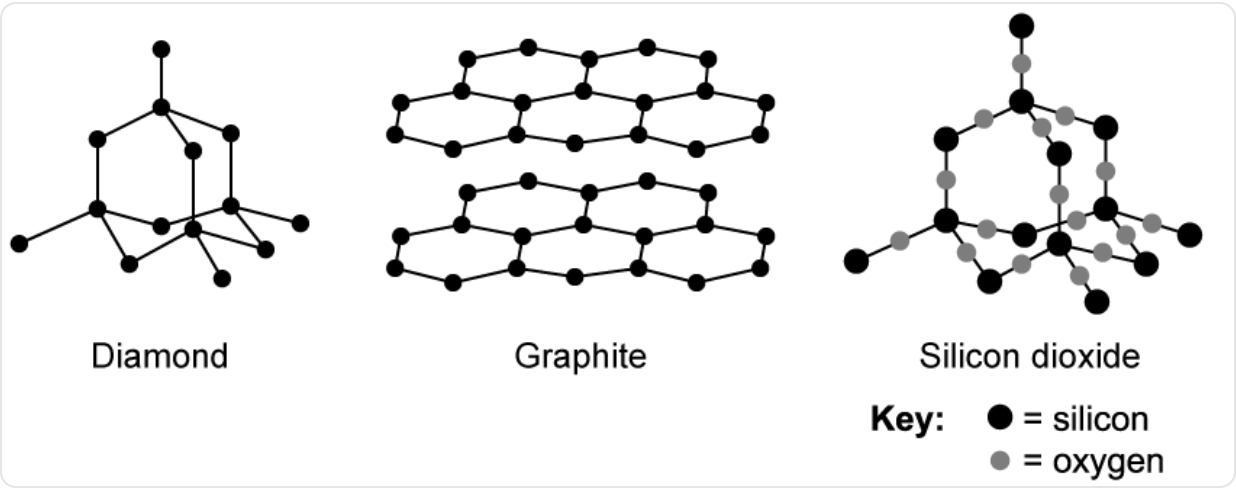
- A) The products have more energy than the reactants in an exothermic reaction
- B) The products have more energy than the reactants in an endothermic reaction
- C) The activation energy for an exothermic reaction is always larger than the activation energy for an endothermic reaction
- D) Most reactions are endothermic

Question 69 (1 mark)

(a)

[1]

The diagram shows the structures of three substances that have giant covalent structures.



Which does **not** provide the correct explanation of the property of a substance?

	Substance	Property	Explanation
A	graphite	conducts electricity	its ions are free to move through the layers
B	diamond	hard	each carbon atom is covalently bonded to four other carbon atoms
C	silicon dioxide	high melting point	it has many strong covalent bonds which require a large amount of energy to break
D	graphite	soft	its layers can slide over each other

.....

.....

.....

Question 70 (1 mark)

(a)

[1]

A reactivity series can be deduced by observing the reactions of metals with water and sulfuric acid.

Four metals were added to water and sulfuric acid and the results recorded below.

Metal	Reaction with water	Reaction with dilute sulfuric acid
W	no reaction	no reaction
X	very slow reaction	reacts quickly
Y	no reaction	reacts slowly
Z	reacts quickly	reacts violently

What is the order of reactivity starting with the least reactive?

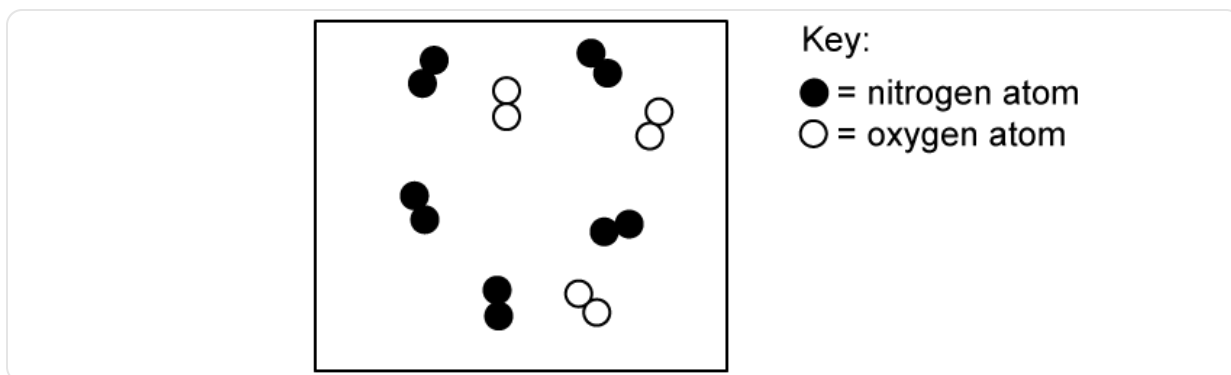
- A) W, Y, X, Z
- B) Y, W, X, Z
- C) W, Y, Z, X
- D) Z, X, Y, W

Question 71 (1 mark)

(a)

[1]

The diagram below represents a sample of air:



What correctly describes this sample of air?

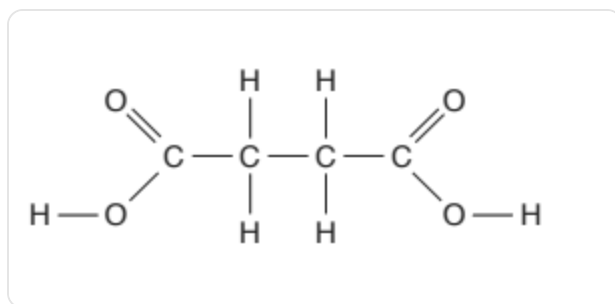
- A) an element
- B) a compound
- C) a mixture of elements
- D) a mixture of compounds

Question 72 (1 mark)

(a)

[1]

The diagram shows the displayed formula of succinic acid.



What is the **empirical formula** for this compound?

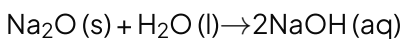
- A) $\text{C}_4\text{H}_6\text{O}_4$
- B) CH_2O
- C) $\text{C}_2\text{H}_3\text{O}_2$
- D) CHO

Question 73 (1 mark)

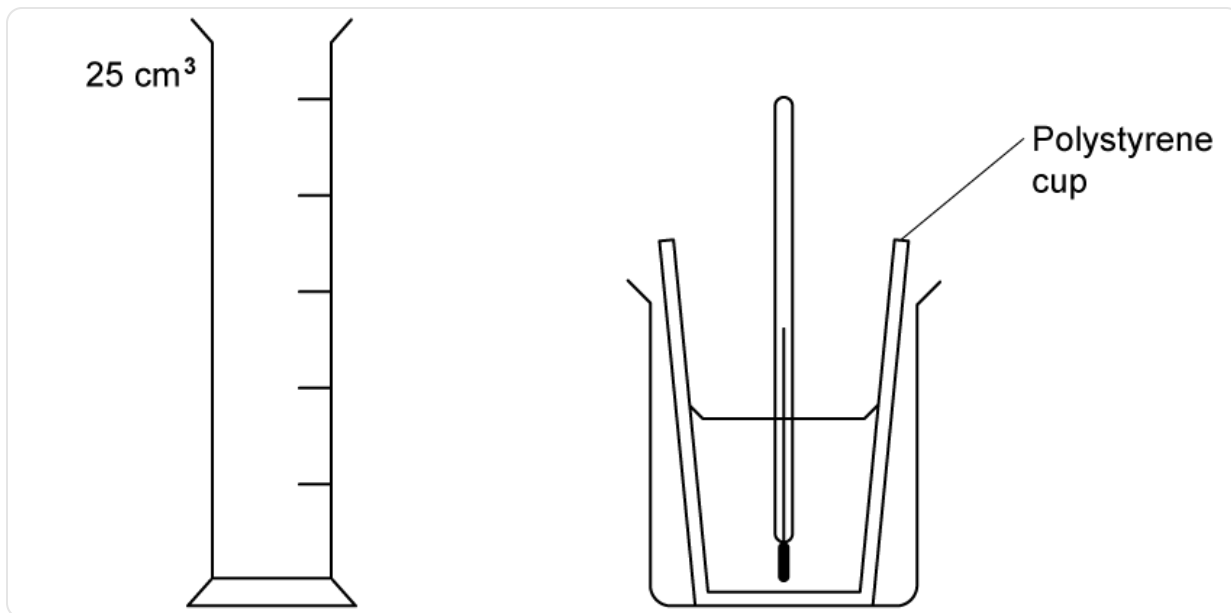
(a)

[1]

When solid sodium oxide is added to water, an exothermic reaction occurs:



A student uses the apparatus below to measure the temperature change of the reaction:



The student records the following results for the reaction:

mass of sodium oxide used = 3.0 g

volume of water used = 50 cm³

starting temperature of mixture = 19.5 °C

maximum temperature of mixture = 30.5 °C

What is the heat energy, Q , released in this experiment, in J?

$c = 4.2 \text{ J/g/}^\circ\text{C}$

mass of 1 cm³ of water = 1 g

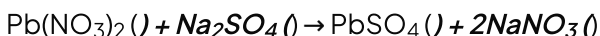
- A) $Q = 2,310 \text{ J}$
- B) $Q = 2,450 \text{ J}$
- C) $Q = 139 \text{ J}$
- D) $Q = 126 \text{ J}$

Question 74 (1 mark)

(a)

[1]

The chemical equation for the preparation of lead(II) sulfate is written below.



What are the state symbols for each substance in this reaction?

- A) aq, aq, s, aq
- B) aq, aq, s, s
- C) s, aq, aq, s
- D) s, aq, s, aq

Question 75 (1 mark)

(a)

[1]

How many moles are present in 9 kg of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$?

[M_r of $\text{C}_6\text{H}_{12}\text{O}_6 = 180$]

- A) 0.02 mol
- B) 0.05 mol
- C) 20 mol
- D) 50 mol

Question 76 (1 mark)

(a)

[1]

Ammonium ions, NH_4^+ , in a solution can be identified by a chemical test.

Which statement is **incorrect** about the test for the ammonium ion, NH_4^+ ?

- A) Add sodium hydroxide solution
- B) A positive test produces a pungent-smelling gas
- C) Use dry red litmus paper to test the gas produced
- D) The litmus paper should turn blue if NH_4^+ ions were present

Question 77 (1 mark)

(a)

[1]

Astatine is found below iodine in Group 7 of the Periodic Table.

What is its formula and state at room temperature?

- A) Formula: At State at room temperature: gas
- B) Formula: At State at room temperature: solid
- C) Formula: At_2 State at room temperature: liquid
- D) Formula: At_2 State at room temperature: solid

Question 78 (1 mark)

(a)

[1]

Which diagram shows the electronic configuration of a nitrogen atom?

.....

.....

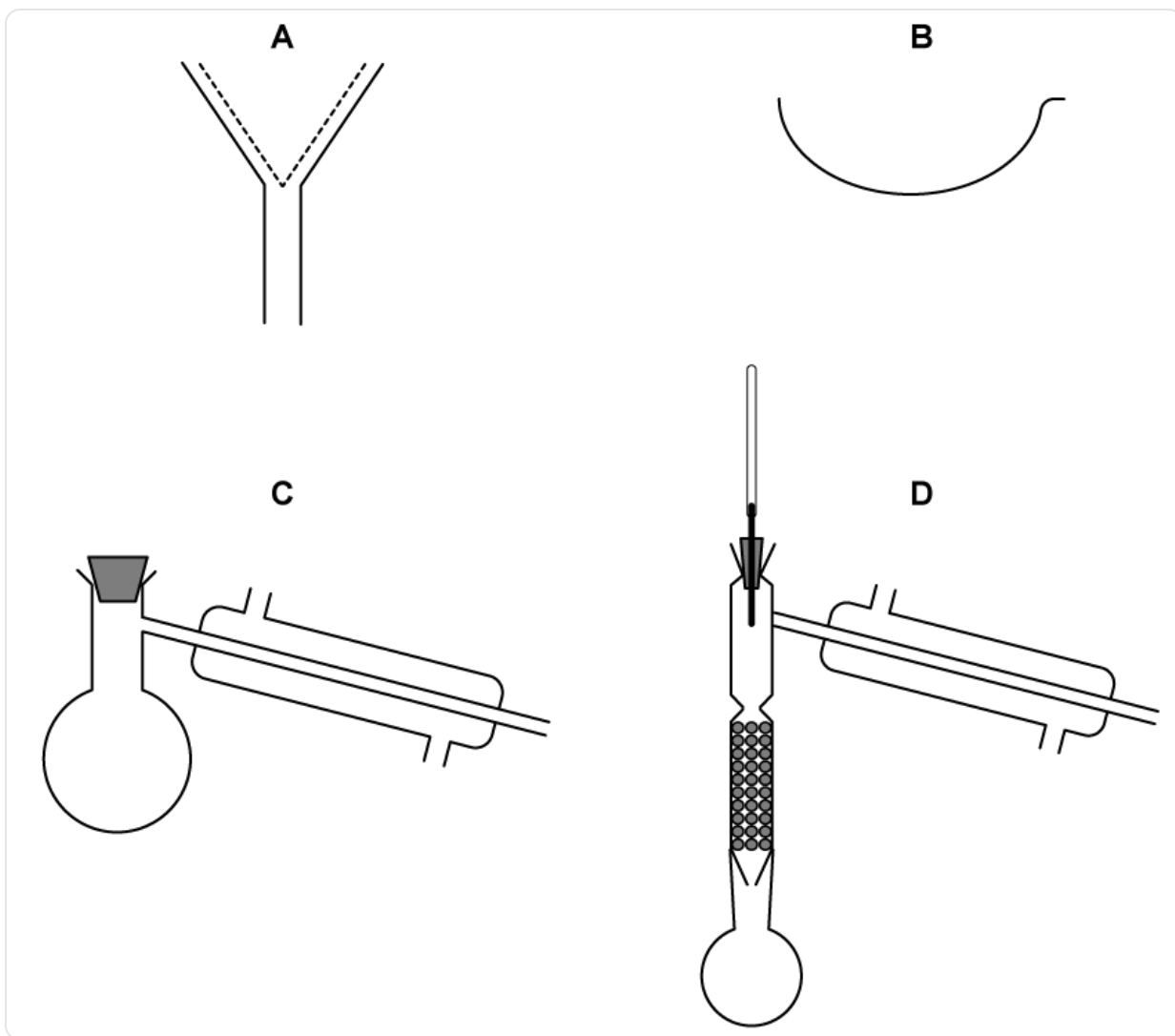
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Question 79 (1 mark)

(a)

[1]

The diagram shows four pieces of apparatus for separating mixtures:



Which one would be most suitable for separating a mixture of alcohol and water?

.....

.....

.....

Question 80 (1 mark)

(a)

[1]

Ethanol is produced by fermentation.

Which of the following conditions is **not** required to produce ethanol by fermentation?

- A) A temperature of between 15 ^oC and 35 ^oC
- B) The addition of yeast
- C) The presence of oxygen
- D) Sugar or starch dissolved in water

Question 81 (1 mark)

(a)

[1]

Metals exist in giant metallic structures which have many strong metallic bonds.

Which of these properties of metals can be explained by this feature of their structure?

- A) Good conductors of electricity
- B) Malleable and ductile
- C) High melting point
- D) High density

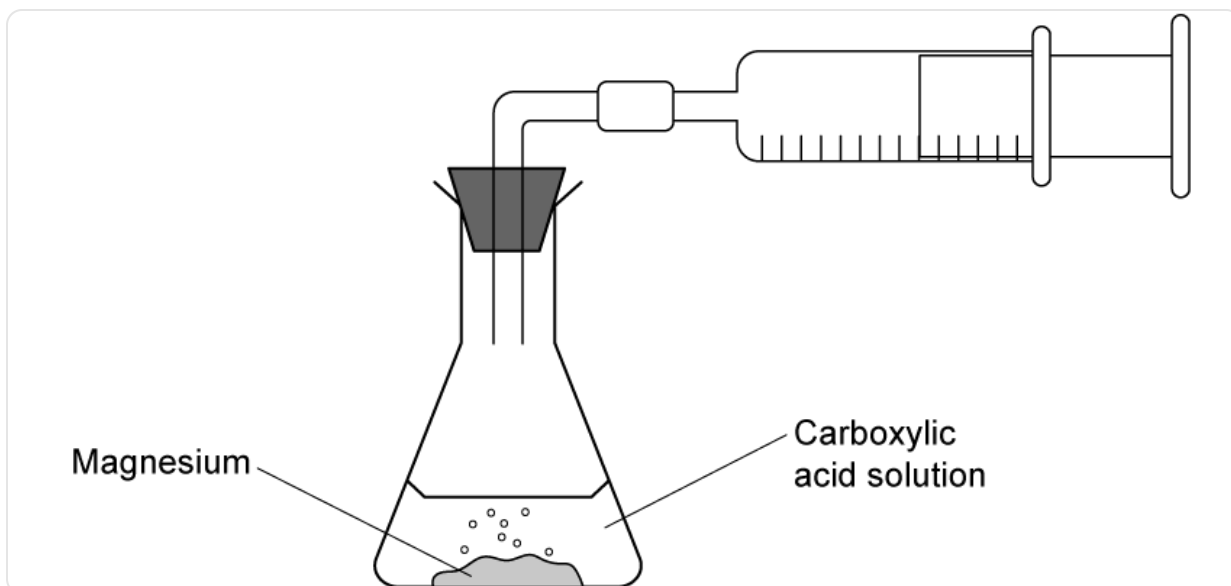
Question 82 (1 mark)

(a)

[1]

Solutions of carboxylic acids react with magnesium to form a salt and hydrogen gas.

Using the apparatus and method below, a student investigates how the time to collect 20 cm³ of hydrogen gas varies with different carboxylic acid solutions.



Method:

- pour the carboxylic acid solution into a conical flask
- add some magnesium into the carboxylic acid solution
- immediately connect the gas syringe and start a timer
- record the time taken to collect 20 cm³ of hydrogen gas
- repeat for different carboxylic acid solutions

Which of the following is **not** a variable that should be controlled?

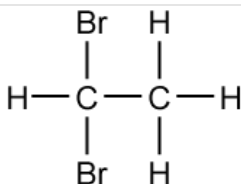
- A) The mass of magnesium
- B) The volume of acid
- C) The type of acid
- D) The concentration of acid

Question 83 (1 mark)

(a)

[1]

The displayed formulae of an organic compound is shown below.



What is the empirical formula of this compound?

- A) CHBr
- B) CH₂Br
- C) C₂H₄Br₂
- D) CH₄Br

Question 84 (1 mark)

(a)

[1]

A student was preparing the insoluble salt lead(II) sulfate from solutions of lead(II) nitrate and potassium sulfate.

Why could the student not use lead(II) carbonate to prepare this salt?

- A) It has a high melting point
- B) It is insoluble in water
- C) it is toxic
- D) It is flammable

Question 85 (1 mark)

(a)

[1]

What is the correct chemical formula of butane?

- A) C₃H₆
- B) C₃H₈
- C) C₄H₈
- D) C₄H₁₀

Question 86 (1 mark)

(a)

[1]

A student wants to investigate the effect of surface area on the rate of reaction between marble chips and dilute acid.

What pieces of apparatus could be used for this investigation?

- A) Burette, beaker, pipette
- B) Ruler, thermometer, test tube
- C) Measuring cylinder, delivery tube, stopwatch
- D) Filter funnel, evaporating basin, Bunsen burner

Question 87 (1 mark)

(a)

[1]

Methanoic acid and ethanol react in the presence of an acid to form an ester.

Name the ester that will be formed.

- A) Methyl methanoate
- B) Methyl ethanoate
- C) Ethyl methanoate
- D) Ethyl ethanoate

Question 88 (1 mark)

(a)

[1]

When ethanol is heated with potassium dichromate(VI) and one other reagent, a reaction occurs.

Which statement about the reaction is correct?

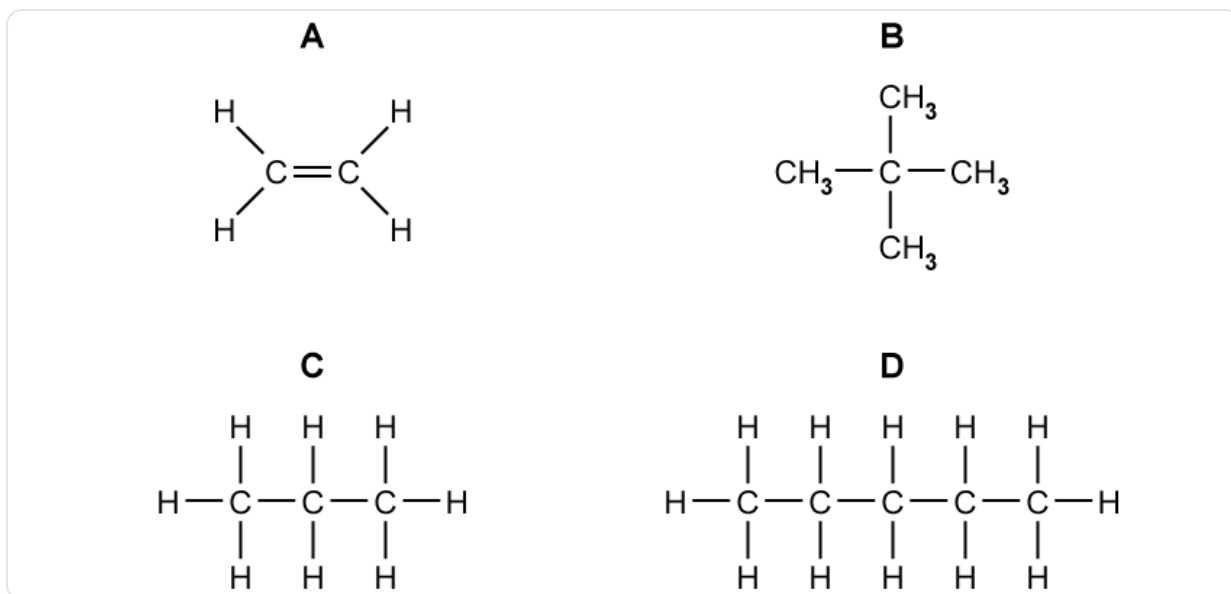
- A) The colour change that occurs during the reaction is from green to orange
- B) The other reagent is sulfuric acid
- C) The product of the reaction is ethene
- D) This is an example of a displacement reaction

Question 89 (1 mark)

(a)

[1]

The structures of four organic compounds are shown.



Which one does **not** show the displayed formula of the compound?

.....

.....

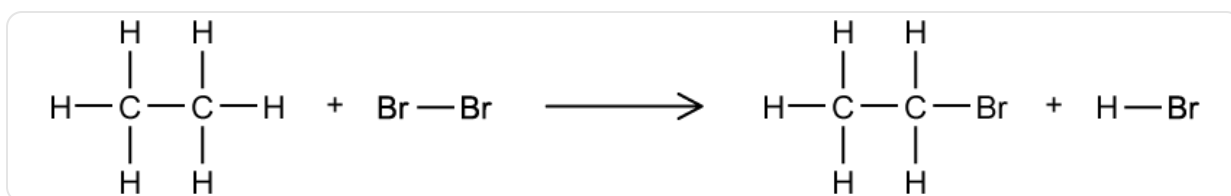
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Question 90 (1 mark)

(a)

[1]

Ethane and bromine can react in the following reaction:



What name is given to this type of reaction?

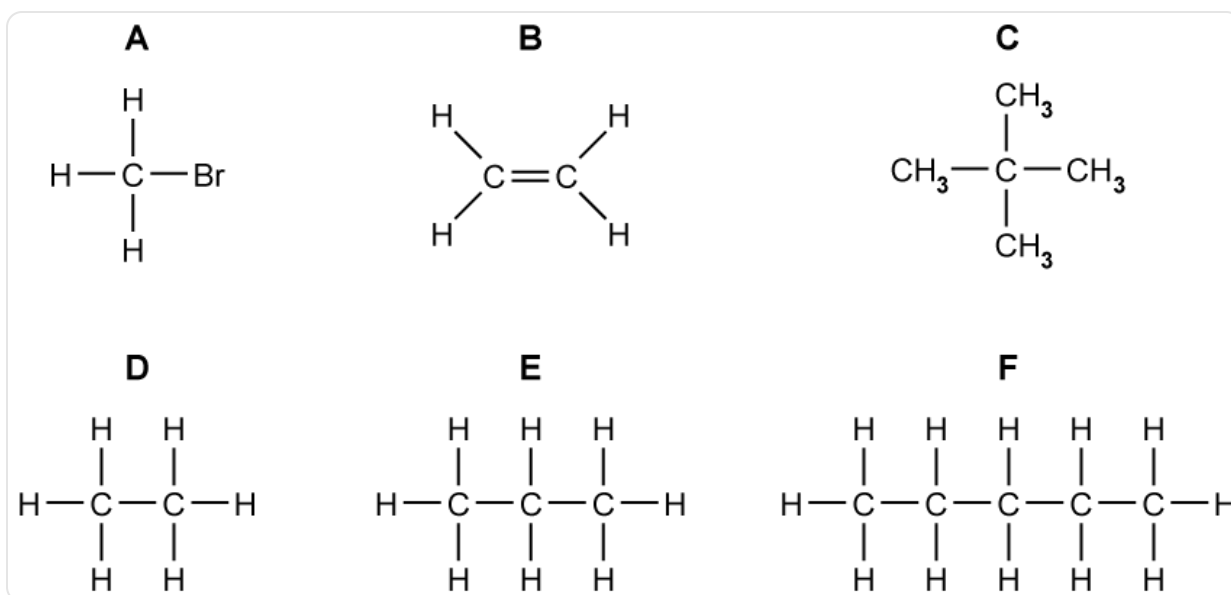
- A) Substitution
- B) Addition
- C) Dehydration
- D) Thermal decomposition

Question 91 (1 mark)

(a)

[1]

The structures of six organic compounds are shown.



Which two compounds are isomers of one another?

- A) B and D
- B) A and C
- C) C and F
- D) C and E

Question 92 (1 mark)

(a)

[1]

The following elements are found in group 0 of the periodic table.

He Ne Ar Kr Xe Rn

What explains their similar chemical and physical properties?

- A) They have the same number of valence electrons
- B) They are all gases
- C) They have the same number of protons and electrons
- D) They have full outer shells

Question 93 (1 mark)

(a)

[1]

Kerosene, bitumen and diesel oil are fractions obtained through the fractional distillation of crude oil.

Which of the following shows the correct use of each fraction?

	kerosene	bitumen	fuel oil
☐ A	fuel for aircraft	surfacing roads	in power stations
☐ B	fuel for trains	in power stations	fuel for aircraft
☐ C	in domestic heating	surfacing roads	in power stations
☐ D	fuel for aircraft	in domestic cooking	in domestic cooking

Question 94 (1 mark)

(a)

[1]

The table below shows the properties of four substances, **W**, **X**, **Y** and **Z**.

Substance	Melting point	Boiling point	Conducts electricity when	
	/°C	/°C		
	solid	liquid		
W	3410	5930	yes	yes
X	801	1413	no	yes
Y	3550	4830	no	no
Z	-91	98	no	no

Use the information in the table to identify the substance that is an ionic compound.

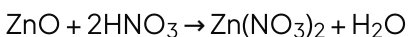
- A) substance **W**
- B) substance **X**
- C) substance **Y**
- D) substance **Z**

Question 95 (1 mark)

(a)

[1]

When zinc oxide reacts with dilute nitric acid, zinc nitrate is produced. The equation for the reaction is:



0.200 mol of nitric acid reacts with excess zinc oxide. A mass of 15.3 g of zinc nitrate is produced. Calculate the percentage yield of zinc nitrate. [M_r of zinc nitrate = 189]

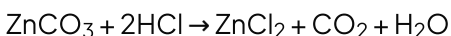
- A) 26%
- B) 37%
- C) 81%
- D) 65%

Question 96 (1 mark)

(a)

[1]

A student wanted to make 15.0 g of zinc chloride. The equation for the reaction is:



What mass of zinc carbonate should the student add to the hydrochloric acid to make 15.0 g of zinc chloride?

[A_r of C = 12; Zn = 65; O = 16; Cl = 35.5; H = 1]

- A) 11.0 g
- B) 13.8 g
- C) 15.0 g
- D) 22.0 g

Question 97 (1 mark)

(a)

[1]

This table gives information about some of the halogens.

Halogen	Physical state at room temperature	Colour
fluorine	gas	
chlorine		pale green
bromine	liquid	red-brown

What is the correct missing physical state and colour?

- A) Physical state at room temperature: gas Colour: colourless
- B) Physical state at room temperature: liquid Colour: colourless
- C) Physical state at room temperature: gas Colour: pale yellow
- D) Physical state at room temperature: liquid Colour: pale yellow

Question 98 (1 mark)

(a)

[1]

A solution containing carbonate ions, CO_3^{2-} , can be reacted with hydrochloric acid, HCl, and a gaseous product collected and tested.

Which is a correct positive test result for a product of this reaction?

- A) A gas is produced which causes cloudiness in limewater
- B) A gas is produced which makes a squeaky pop sound when a lit splint is introduced
- C) A gas is produced which bleaches damp litmus paper
- D) A gas is produced which re-ignites a glowing splint

Question 99 (1 mark)

(a)

[1]

Ethyl ethanoate can be prepared by heating a mixture of ethanoic acid, ethanol and concentrated sulfuric acid.

Which statement about this reaction is **not** correct?

- A) Concentrated sulfuric acid acts as catalyst
- B) The ester distills first due to its low boiling point
- C) The mixture is heated using a water bath not a Bunsen burner
- D) The ethyl ethanoate collected is pure

Question 100 (1 mark)

(a)

[1]

Halon 1301 is a compound used in some fire extinguishers.

Halon 1301 has the percentage composition by mass of 8.05% C, 53.69% Br and 38.26% F.

Calculate the empirical formula of Halon 1301.

[A_r of C = 12; Br = 80; F = 19]

- A) CBrF
- B) CBrF_3
- C) CBr_3F
- D) CBr_2F_2

Question 101 (1 mark)

(a)

[1]

A student adds a few drops of dilute nitric acid followed by a few drops of silver nitrate solution to a test tube containing an unknown colourless solution. A white precipitate is seen.

What ions could be present in the solution?

- A) Ammonium ions
- B) Chloride ions
- C) Sulfate ions
- D) Magnesium ions

Question 102 (1 mark)

(a)

[1]

The reaction between aqueous chlorine and aqueous potassium bromide results in the formation of an orange solution.

Which is the correct statement about this reaction?

- A) Bromine is more reactive than chlorine
- B) Potassium chloride is responsible for the orange colour
- C) Chlorine displaces bromine from potassium bromide
- D) Bromide ions gain electrons from chlorine

Question 103 (1 mark)

(a)

[7]

Which of these is true about the manufacture of ethanol?

- A) Fermentation has a 100% atom economy
B) The hydration of ethene has a 100% atom economy
C) Fermentation produces pure ethanol
D) Alkenes are reactive so the hydration of ethene occurs readily at room temperature

Question 104 (1 mark)

(a)

[7]

Butanoic acid behaves like other acids.

Which is the correct statement about the reactions of butanoic acid?

- A) It reacts with sodium hydroxide in a neutralisation reaction to form a salt and hydrogen gas
- B) In a reaction with calcium carbonate, a salt and water are formed
- C) It reacts with calcium to form a salt and oxygen gas
- D) When it reacts with magnesium carbonate, fizzing is observed

Question 105 (1 mark)

(a)

[7]

Which element, T, Q, R, or M has the electronic configuration 2,8,3?

[illegible]

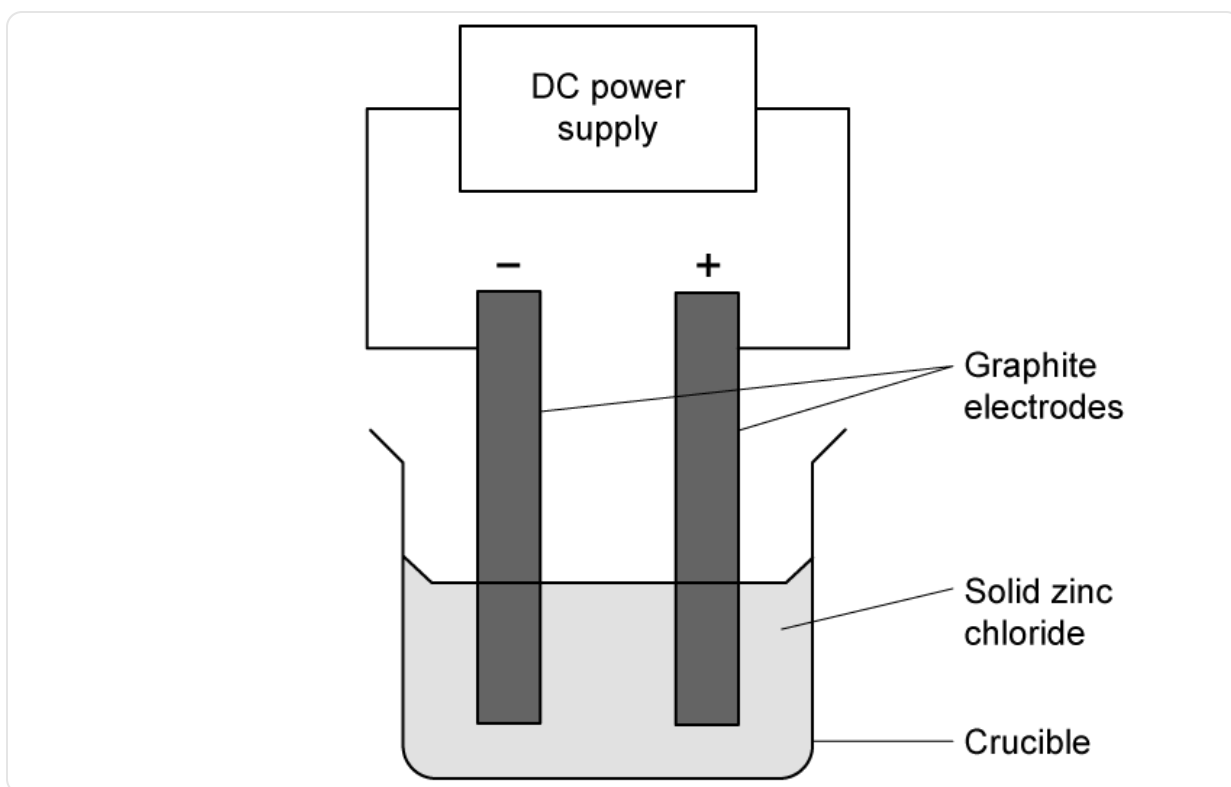
- A) element M
B) element Q
C) element T
D) element R

Question 106 (1 mark)

(a)

[1]

A student was investigating the electrolysis of zinc chloride.



What is the correct reason for electrolysis not working?

- A) Solid zinc chloride has been used
- B) The electrodes don't conduct electricity
- C) The power supply is not connected properly
- D) The crucible is too small to allow ions to flow

Question 107 (1 mark)

(a)

[1]

The table below shows the properties of four substances, **W**, **X**, **Y** and **Z**.

Substance	Melting point	Boiling point	Conducts electricity when	
	/°C	/°C	solid	liquid
W	3410	5930	yes	yes
X	801	1413	no	yes
Y	3550	4830	no	no
Z	-91	98	no	no

Use the information in the table to identify the substance that is a metal.

- A) Substance W
- B) Substance X
- C) Substance Y
- D) Substance Z

Question 108 (1 mark)

(a)

[1]

Incomplete combustion occurs when there is insufficient oxygen for a reaction.

Which are the **two** main substances formed during incomplete combustion, aside from water?

- A) Particulates and carbon monoxide
- B) Soot and sulfur dioxide
- C) Carbon dioxide and carbon monoxide
- D) Oxides of nitrogen and sulfur dioxide

Question 109 (1 mark)

(a)

[1]

What is the empirical formula of butanoic acid?

- A) C_3H_7COOH
- B) $C_4H_8O_2$
- C) C_2H_4O
- D) $C_2H_4O_2$

Question 110 (1 mark)

(a)

[1]

Which of the following dot-and-cross diagrams shows the correct arrangement of electrons in lithium sulfide?

.....

.....

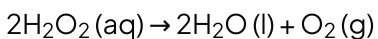
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Question 111 (1 mark)

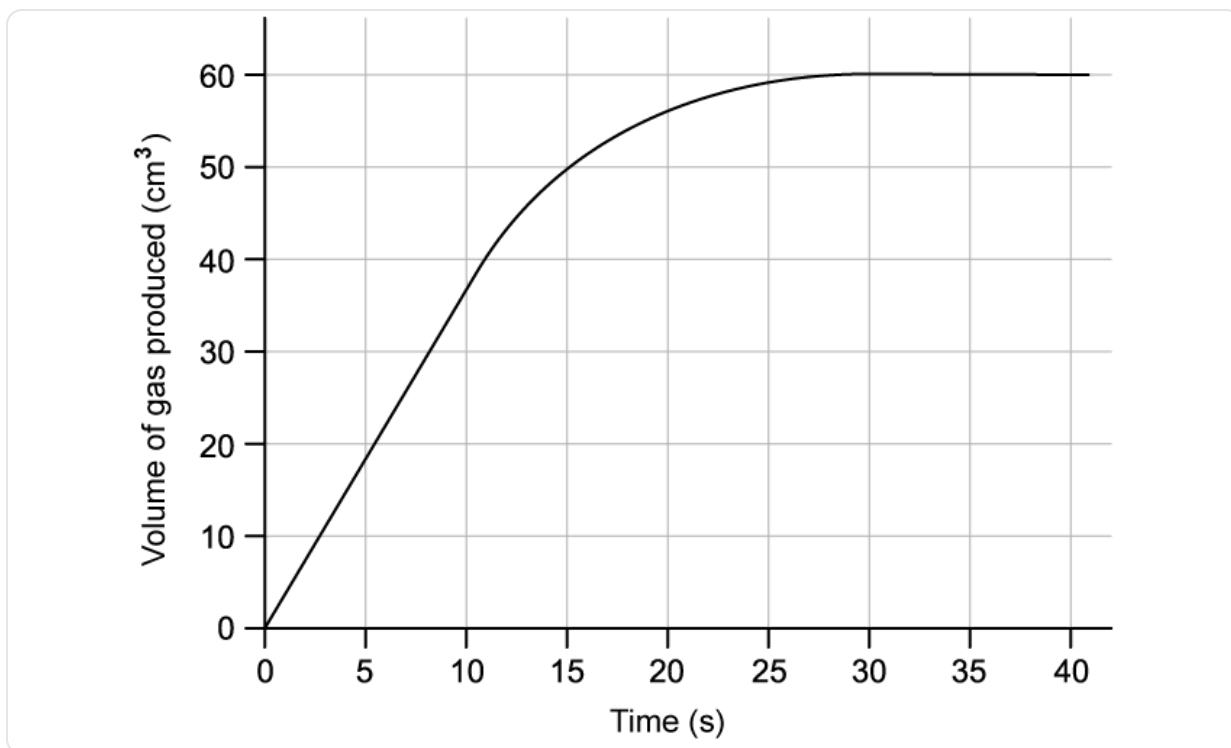
(a)

[1]

Hydrogen peroxide, H_2O_2 , quickly decomposes to produce water and oxygen when a catalyst is added to the solution:



The volume of gas produced was measured and is shown in the graph below:



Which statement is **not** true for this reaction?

- A) The reaction finishes after 30 seconds
- B) The rate of reaction speeds up during the reaction
- C) The final volume of gas produced is 60 cm^3
- D) The concentration of the hydrogen peroxide decreases during the reaction

Question 112 (1 mark)

(a)

[1]

What is the mass of 0.25 moles of hydrated copper(II) sulfate, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$?

Relative formula mass (M_r): $\text{H}_2\text{O} = 18$ $\text{CuSO}_4 = 160$

- A) 40.0 g
- B) 44.5 g
- C) 62.5 g
- D) 1000 g

Question 113 (1 mark)

(a)

[1]

Silicon dioxide, SiO_2 , and silicon(IV) chloride, SiCl_4 , are both covalently bonded compounds.

The table below shows the melting point and boiling point of these compounds.

Compound	Melting point / °C	Boiling point / °C
Silicon dioxide	1710	2230
Silicon(IV) chloride	-69	58

Which statement is **not** correct about the two compounds?

- A) at room temperature silicon dioxide is a solid and silicon(IV) chloride is a liquid
- B) silicon dioxide has a giant covalent structure
- C) silicon(IV) chloride has a lower melting point as it has fewer covalent bonds to overcome
- D) silicon(IV) chloride has a simple molecular structure

Question 114 (1 mark)

(a)

[1]

Why can esters be used as food flavourings?

- A) They are volatile
- B) They have distinctive colours
- C) They have fragrant odours
- D) They are insoluble in water

Question 115 (1 mark)

(a)

[1]

A reaction took place between iron sulfate and magnesium.

iron(II) sulfate + magnesium $\rightarrow$ magnesium sulfate + iron

Which species has been oxidised?

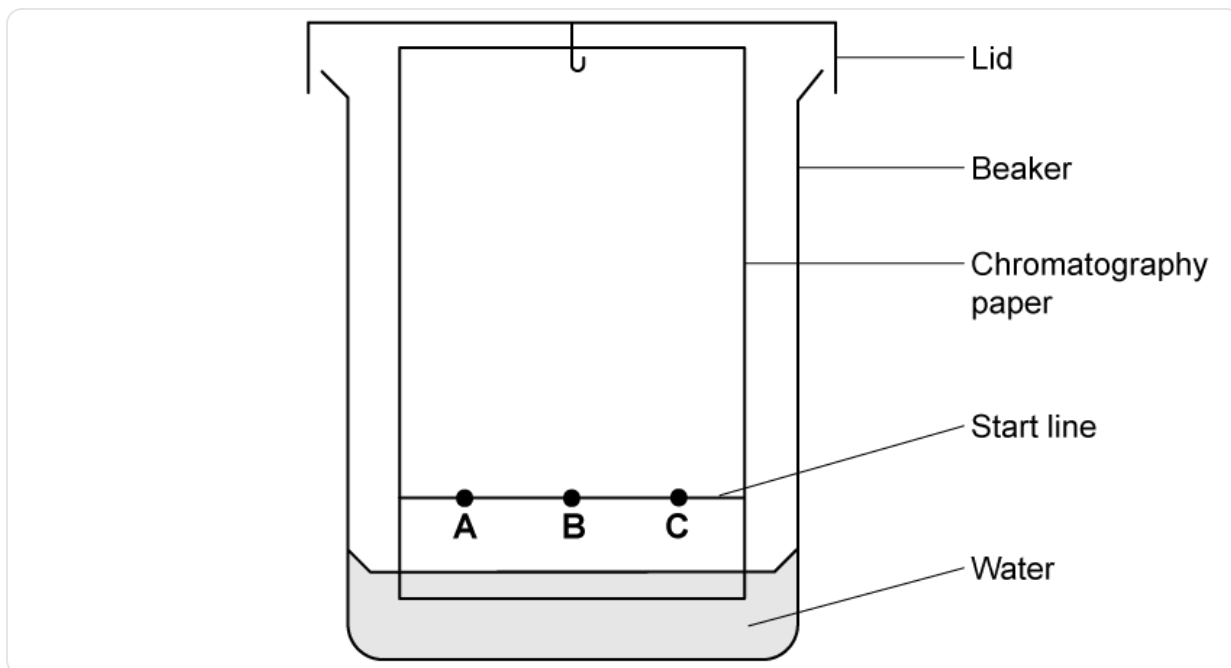
- A) Iron
- B) Magnesium
- C) Sulfur
- D) None of the above

Question 116 (1 mark)

(a)

[1]

A student set up the following apparatus to investigate the colours in three different food colourings.



What are the stationary and mobile phases for this experiment?

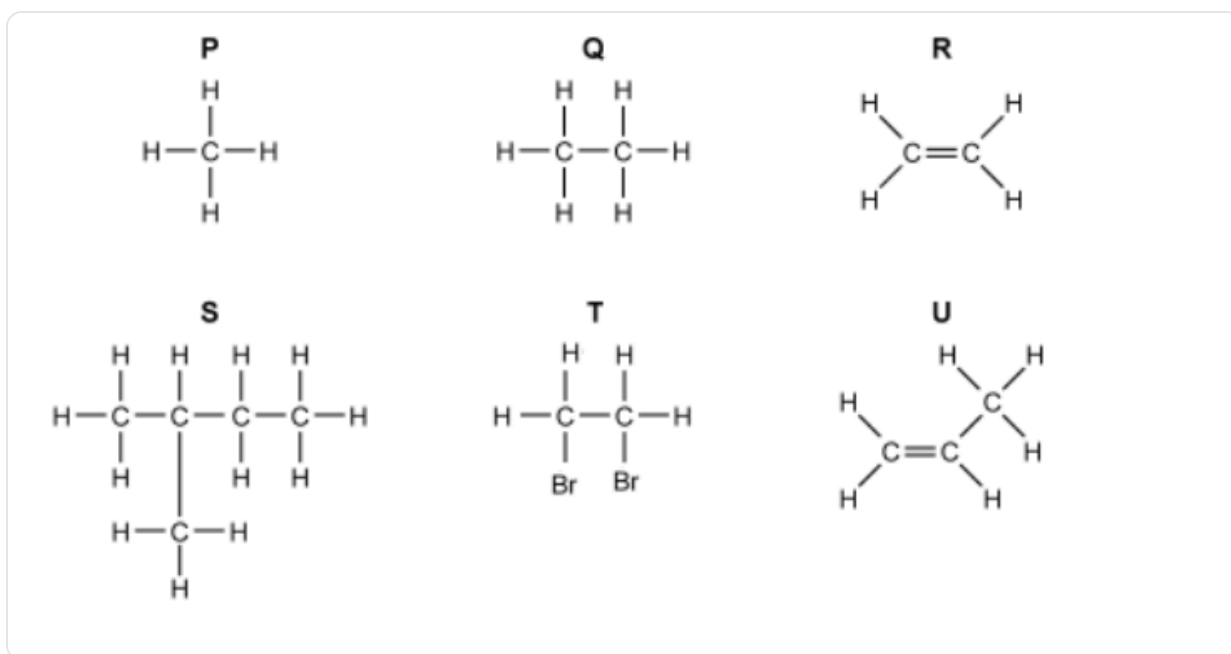
- A) stationary: chromatography papermobile: water
- B) stationary: beakermobile: start line
- C) stationary: watermobile: food colourings
- D) stationary: start linemobile: chromatography paper

Question 117 (1 mark)

(a)

[1]

The table shows the displayed formulae of some organic compounds.



Which two are **not** classed as alkanes?

- A) R and S
- B) S and T
- C) P and Q
- D) R and T

Question 118 (3 marks)

(a) suggest

[3]

Refractory bricks are special bricks used in lining furnaces, kilns, and fireplaces.

Magnesium oxide can be used to make refractory bricks.

Suggest why magnesium oxide is suitable for refractory bricks.

Give a reason, referring to the structure and bonding.

.....

.....

.....

Question 119 (3 marks)

(a) explain

[3]

Explain, in terms of electron configuration, why fluorine is more reactive than chlorine.

.....

.....

.....

Question 120 (4 marks)

(a) give

[7]

The diagram shows part of the Periodic Table, with elements represented by the letters L, M, Q, R and T.

The letters in the diagram represent elements but are not their chemical symbols.

Diagram illustrating a 1D lattice with 12 sites. The sites are labeled 1 through 12. The particles are distributed as follows:

- Site 1: L
- Site 2: M
- Site 3: (empty)
- Site 4: (empty)
- Site 5: (empty)
- Site 6: (empty)
- Site 7: (empty)
- Site 8: Q
- Site 9: R
- Site 10: T
- Site 11: (empty)
- Site 12: 0

Give the letter from the diagram that represents a noble gas.

(b) state

[7]

Elements L and M are in the same group.

State why they have similar chemical reactions.

(c) explain

[2]

An atom of element Q has 31 protons.

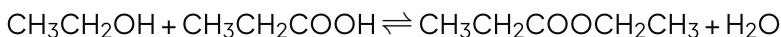
Use this information to explain how you can determine the number of protons in an atom of element R.

Question 121 (4 marks)

(a) multiple

[3]

An ester is formed in the following dynamic equilibrium reaction between ethanol and propanoic acid:



i) Name the ester produced.

[1]

ii) State two characteristics of a dynamic equilibrium.

[2]

.....

.....

.....

(b)

[1]

Which structure represents part of an ester molecule but **not** part of a carboxylic acid molecule?

.....

.....

.....

Question 122 (4 marks)

(a) complete

[4]

The table contains statements about elements in the Periodic Table.

Complete the missing information.

[4]

Statement
The name of the element with atomic number 18.
The name of the element with relative atomic mass 7.
The name of the element in Group 3, Period 4.
The electronic configuration of the element phosphorus.

.....

.....

.....

Question 123 (4 marks)

(a)

[1]

The table shows some properties of four substances, P, Q, R and S.

Substance	Melting Point /°C	Boiling Point /°C	Conducts electricity when	
solid	liquid			
P	3410	5930	yes	yes
Q	734	1435	no	yes
R	-95	69	no	no
S	2507	3900	no	no

Use the information in the table to answer the following questions. You may use each letter once, more than once or not at all.

Choose a substance that is a solid at 3000 °C.

- A) substance **P**
- B) substance **Q**
- C) substance **R**
- D) substance **S**

(b)

[1]

Choose a substance that is a liquid at 25 °C.

- A) substance **P**
- B) substance **Q**
- C) substance **R**
- D) substance **S**

(c)

[1]

Choose a substance that is an ionic compound.

- A) substance **P**
- B) substance **Q**
- C) substance **R**
- D) substance **S**

(d)

[1]

Choose a substance that is a metal.

- A) substance **P**
- B) substance **Q**
- C) substance **R**
- D) substance **S**

Question 124 (4 marks)

(a)

[1]

Sulfuric acid can be used to make a soluble salt when reacted with a base.

All acids contain the same ion. What is the formula of this ion?

- A) H^{+}
- B) OH^{-}
- C) S^{2-}
- D) Na^{+}

(b)

[1]

Sulfuric acid reacts with copper(II) oxide to form a salt. What is the name of the salt formed?

- A) Copper(II) chloride
- B) Copper(II) nitrate
- C) Copper(II) sulfate
- D) Copper(II) citrate

(c) write

[1]

The steps below show the method the student carried out to make the salt using the reactants in part b).

The steps shown are not in the correct order.

step J	Leave the solution to cool
step K	Filter the mixture and transfer the filtrate to an evaporating basin
step L	Add copper(II) oxide in excess
step M	Heat the acid
step N	Evaporate the water until crystals appear

Write the steps in the correct order.

Some have been completed for you.

first step	last step
M	J

(d) explain

[1]

Explain why the copper(II) oxide was added in excess to the sulfuric acid.

Question 125 (4 marks)

(a)

[1]

Which best describes a covalent bond?

- A) Sea of delocalised electrons and positive metal ions
- B) Attraction between positive and negative ions
- C) Shared pair of electrons between two non-metal atoms
- D) Weak intermolecular forces

(b)

[1]

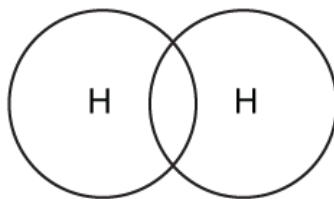
Which of the following substances will bond covalently?

- A) Sodium oxide
- B) Sodium
- C) Sodium bromide
- D) Oxygen

(c) complete

[1]

Hydrogen is an atom which has one electron in the outermost shell. Complete the dot-and-cross diagram to show the bonding in a hydrogen molecule.



(d) why

[1]

Hydrogen is a gas at room temperature.

Why is hydrogen a gas at room temperature?

Tick () **one** box

☐ The intermolecular forces are strong

☐ The bonding in hydrogen is weak

☐ The intermolecular forces are weak

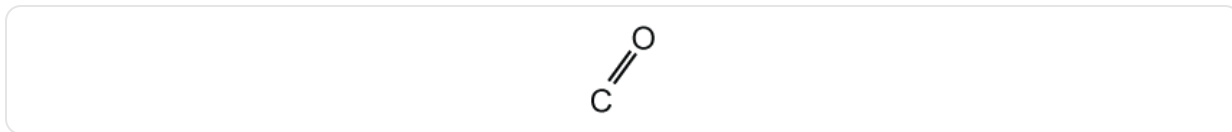
Question 126 (4 marks)

(a) complete

[1]

Butanoic acid belongs to the homologous series carboxylic acids.

Complete the structure of butanoic showing all bonds.



(b) explain

[1]

Explain why butanoic acid is a saturated compound.

(c)

[1]

Butanoic acid can react with magnesium carbonate to form a salt, carbon dioxide and water. What type of reaction is this?

- A) thermal decomposition
- B) neutralisation
- C) combustion
- D) endothermic

(d) complete

[1]

Butanoic acid can also react with a metal.

Complete the word equation for this reaction.

_____ + butanoic acid → sodium butanoate + _____

Question 127 (5 marks)

(a) what [1]

The reactions of metals with water and with dilute sulfuric acid can be used to determine the order of reactivity of the metals.

The table shows the reactions of four metals, W, X, Y and Z, with water and with dilute sulfuric acid.

Metal	Reaction with water	Reaction with dilute sulfuric acid
W	no reaction	no reaction
X	very slow reaction	reacts quickly
Y	no reaction	reacts slowly
Z	reacts quickly	reacts violently

What is the order of reactivity of these metals?

	most reactive			least reactive
A	W	X	Y	Z
B	Z	X	Y	W
C	W	Y	X	Z
D	Z	Y	X	W

(b) state [2]

i) State which metal, W, X, Y or Z, could be copper.

(1)

ii) State which metal, W, X, Y or Z, could be magnesium.

(1)

(c) state [2]

A displacement reaction can also be used to decide the order of reactivity of two metals.

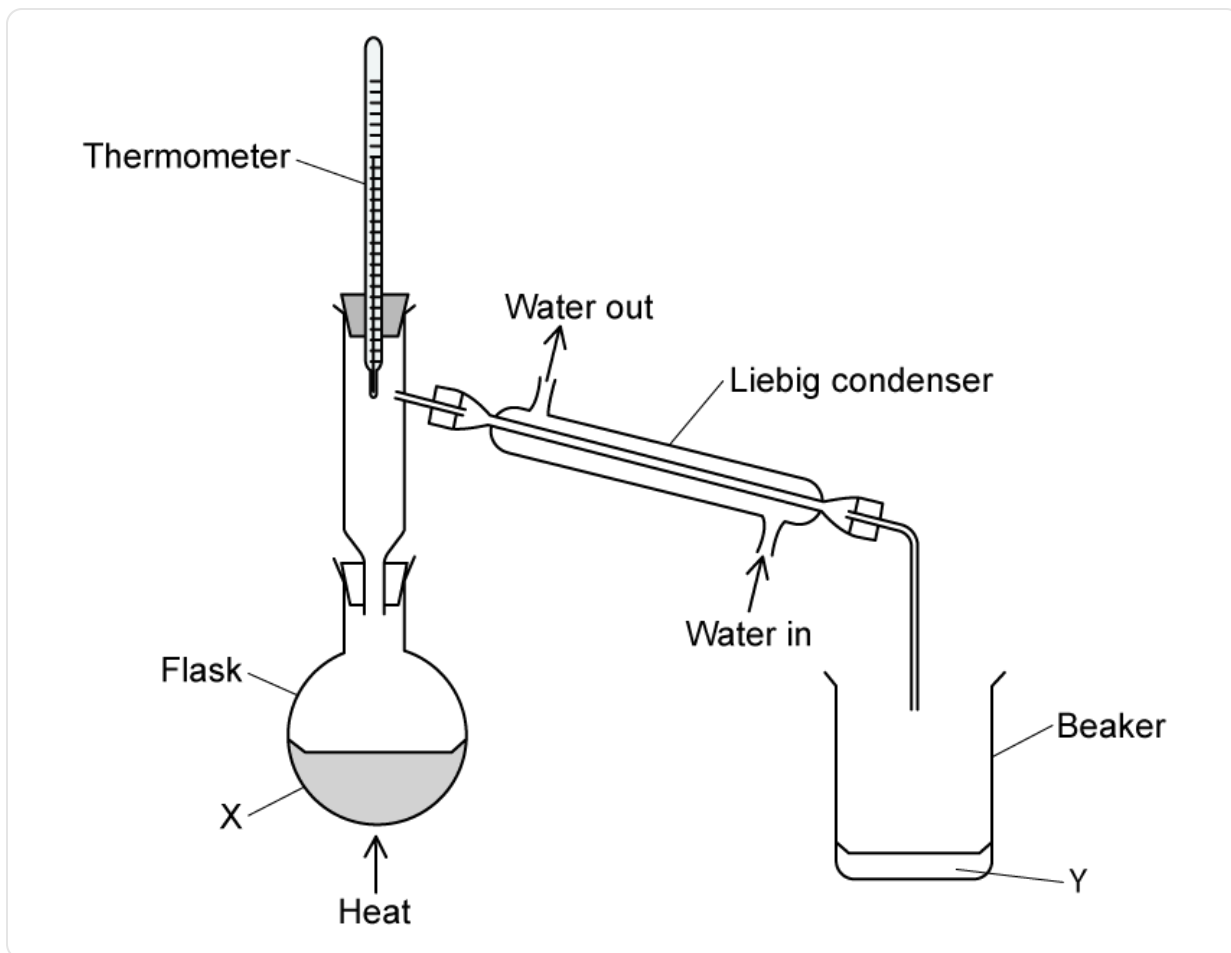
State two observations made when an excess of magnesium powder is added to an aqueous solution of copper(II) sulfate.

Question 128 (5 marks)

(a)

[1]

Ethyl ethanoate can be prepared in the laboratory using the equipment shown in the diagram below.



Which row in the table correctly identifies X and Y.

	X	Y
☐	A ethanoic acid and ethanol	ethyl ethanoate
☐	B ethyl ethanoate	ethanoic acid and ethanol
☐	C ethanoic acid, ethanol and sulfuric acid	ethyl ethanoate
☐	D ethyl ethanoate	ethanoic acid, ethanol and sulfuric acid

(b) tick

[1]

A student carried out a practical to produce ethyl ethanoate using the apparatus shown in part a).

Place a tick (✓) in the box that shows the most suitable method to heat the reaction mixture.

Bunsen burner
spirit burner

Bunsen burner
water bath

(c) give

[1]

The mixture containing ethyl ethanoate may also contain some acid impurities which can be removed by adding sodium carbonate solution to the mixture.

Give one observation the student would make when sodium carbonate was added to the mixture if acid impurities were present.

(d) give

[2]

Propyl butanoate is another ester that has a fruity smell is another ester.

Give the name of the carboxylic acid and alcohol that would be required to produce propyl butanoate.

Carboxylic acid	
Alcohol	

Question 129 (5 marks)

(a) name

[2]

Two substances are needed to cause iron to rust. Name these two substances.

1.....

2.....

.....

.....

.....

(b) give

[3]

The box gives the names of some substances.

calcium	copper	gold
iodine	methane	zinc

Use words from the box to answer these questions.

i) Give the name of a non-metallic element.

(1)

ii) Give the name of a compound.

(1)

iii) Give the name of the metal that is lowest in the reactivity series.

(1)

.....

.....

.....

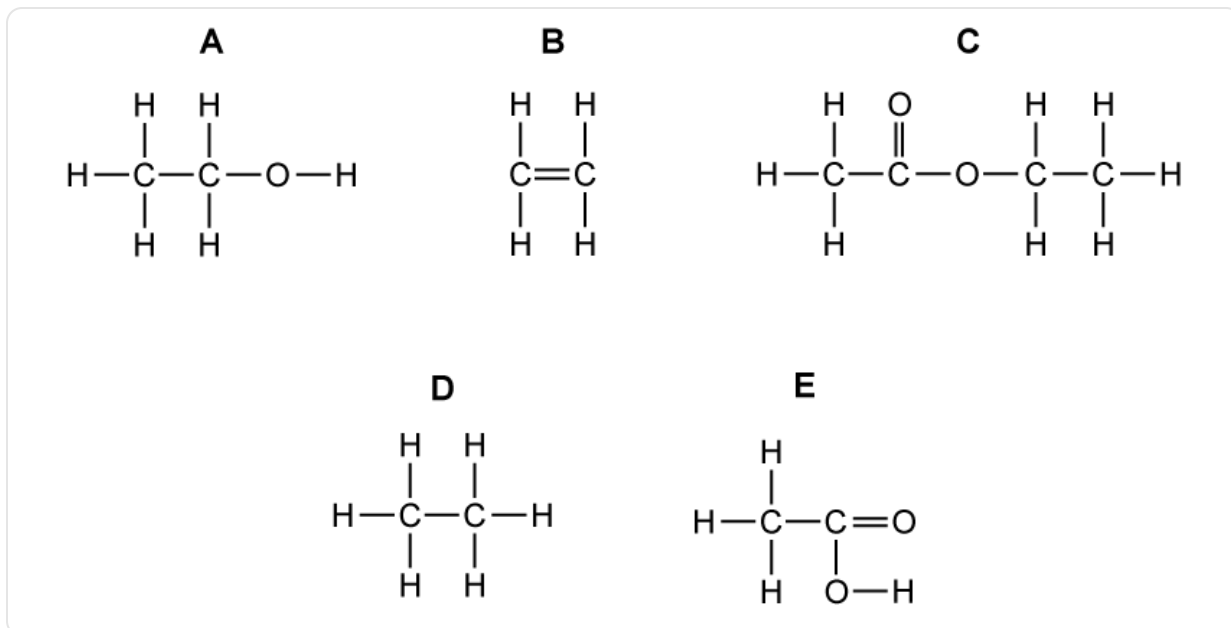
Question 130 (5 marks)

(a) name

[1]

This question is about the reactions of organic compounds **A** – **E** shown in **Figure 1**.

Figure 1



Name compound **C**.

.....

.....

(b) name

[2]

Name the two products formed when **E** reacts with sodium.

.....

.....

(c) which

[1]

Which compound reacts with compound **A** to produce compound **C**?

.....

.....

(d) which

[1]

Which compound dissolves in water to produce a neutral solution?

.....

.....

Question 131 (5 marks)

(a)

[1]

In 1937, an airship full of hydrogen gas flew from Germany to America.



(Source: © Michael Rosskothen/Shutterstock)

Which property of hydrogen makes it a suitable gas to use in an airship?

- A) colourless
- B) insoluble in water
- C) low density
- D) no smell

(b) explain

[2]

Explain why helium is now used in airships instead of hydrogen.

.....

.....

.....

(c) multiple

[2]

Hydrogen is used to manufacture ammonia, NH_3

Hydrogen is reacted with nitrogen using an iron catalyst.

i) Give a chemical equation for this reaction.

(1)

ii) State why a catalyst is used in this reaction.

(1)

.....

.....

.....

Question 132 (5 marks)

(a) calculate

[3]

Compound X is a hydrocarbon with an M_r of 70 that contains 85.7% by mass of carbon. When bromine water is added to compound X it remains orange.

Calculate the molecular formula of compound X

.....

.....

.....

(b) determine

[1]

Using the information in part a), determine the structure of compound X.

.....

.....

.....

(c) write

[1]

Compound X reacts with chlorine in the presence of ultra-violet light. Write the equation for this reaction.

.....

.....

.....

Question 133 (5 marks)

(a)

[1]

A student is given a mixture of salt solution and sand. She wants to obtain pure water from the mixture.

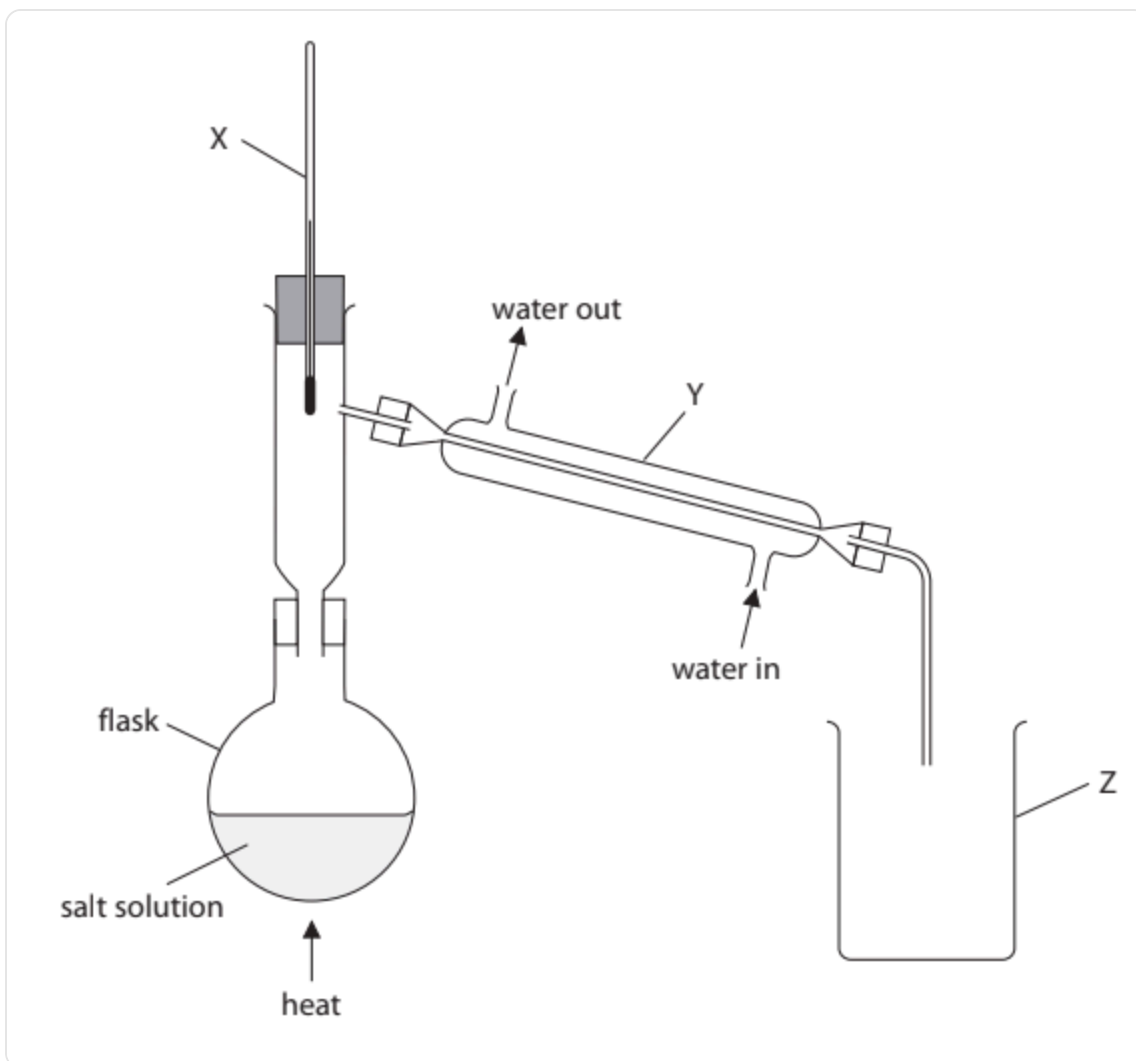
Which method should the student use to separate the sand from the salt solution?

- A) crystallisation
- B) filtration
- C) fractional distillation
- D) simple distillation

(b) multiple

[4]

The student then uses this apparatus to obtain pure water from the salt solution.



i) Name the pieces of apparatus labelled X, Y and Z.

(3)

X

Y

Z

ii) State what remains in the flask when the separation is complete.

(1)

.....

.....

.....

Question 134 (5 marks)

(a)

[1]

Aluminium is an element found in Group 3 of the Periodic Table.

Use your Periodic Table to deduce the number of protons, neutrons and electrons found in an atom of aluminium.

		protons	neutrons	electrons
☐	A	13	27	13
☐	B	13	27	14
☐	C	27	14	14
☐	D	13	14	13

.....

.....

(b) give

[2]

Give two physical properties of aluminium.

.....

.....

(c)

[1]

Duralumin is an alloy of aluminium and copper. What type of substance is an alloy?

- A) Atom
- B) Element
- C) Compound
- D) Mixture

(d) calculate

[1]

Aluminium is extracted from bauxite, a mixture containing aluminium oxide.

One sample of bauxite contains 17% aluminium. Calculate the percentage of substances other than aluminium found in bauxite.

.....

.....

.....

(a) give

The diagram shows the positions of some elements in the Periodic Table.

Use symbols from this table to answer these questions.

Each symbol may be used once, more than once or not at all.

- i) Give the symbol of a metal.

(7)

- ii) Give the symbol of a noble gas.

(7)

- iii) Give the symbol of a liquid at room temperature.

(7)

- iv) Give the symbols of the two elements in Period 3

(7)

..... and

- (b) deduce

[7]

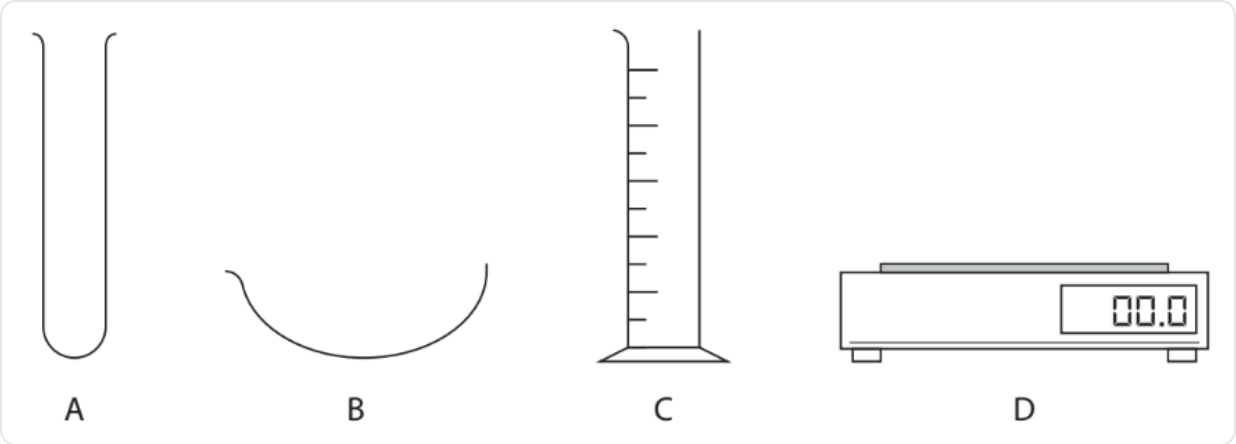
Deduce the electronic configuration of Na.

Question 136 (5 marks)

(a) complete

[4]

The diagram shows some pieces of apparatus.



Complete the table by giving the name of each piece of apparatus.

Letter	Name
A	
B	
C	
D	

(b)

[1]

Which piece of apparatus can be used to measure the volume of a liquid?

Question 137 (5 marks)

(a) name

[1]

This question is about elements, compounds and mixtures.

Name the element that burns with a lilac flame.

.....

.....

(b) name

[1]

Name the technique used to separate the mixture of colours in black ink.

.....

.....

(c) identify

[3]

Indented content here. The box gives the names of some substances.

air	bromine	magnesium	neon	sodium chloride	sulfur
-----	---------	-----------	------	-----------------	--------

i) Choose substances from the box to answer these questions. Identify the compound.

(1)

ii) Identify the mixture.

(1)

iii) Identify the non-metal element that is a solid at room temperature.

(1)

.....

.....

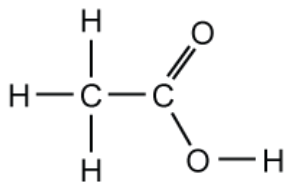
Question 138 (5 marks)

(a)

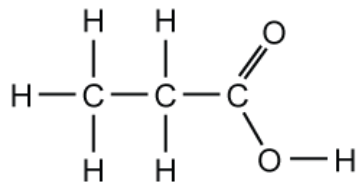
[1]

The structures of two carboxylic acids are shown below.

Carboxylic acid **A**



Carboxylic acid **B**



What is the name of carboxylic acid **B**?

- A) butanoic acid
- B) propanoic acid
- C) ethanoic acid
- D) methanoic acid

(b)

[1]

Give one use of carboxylic acid **A**.

- A) vinegar
- B) fuel
- C) smelling salts
- D) polymer

(c) name

[2]

Both carboxylic acids **A** and **B** can react with metal carbonates to produce a salt and **two** other products. Name the two other products formed.

(d)

[1]

Carboxylic Acids A and B will also react with metals to form hydrogen gas. Which row in the table correctly shows the test and result for this gas?

	Test	Result
A	glowing splint	relights
B	burning splint	squeaky pop sound
C	glowing splint	squeaky pop sound
D	burning splint	flame extinguished

Question 139 (5 marks)

(a) describe

[3]

Describe how long-chain alkanes are converted into short-chain alkanes and alkenes.

.....

.....

.....

(b) explain

[2]

Explain why it is necessary to convert long-chain alkanes into short-chain alkanes and alkenes.

.....

.....

.....

Question 140 (5 marks)

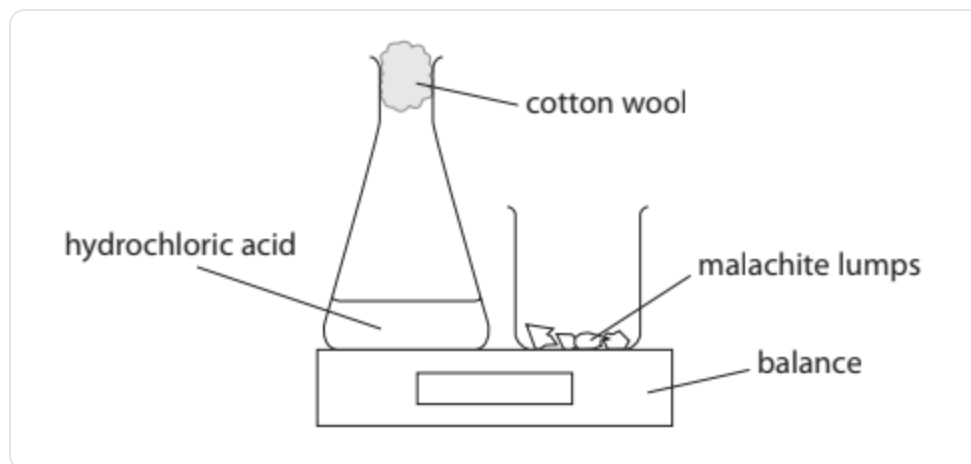
(a) state

[1]

The copper(II) carbonate in the mineral, malachite, reacts with hydrochloric acid according to this equation.



Some students investigate the effect of changing the concentration of acid on the rate of this reaction. The diagram shows the apparatus they use.



This is the method they use:

- set the balance to zero
- add an excess of malachite lumps to the conical flask and replace the cotton wool
- start a timer and record the balance reading after one minute

The experiment is repeated using different concentrations of hydrochloric acid. The mass and number of malachite lumps are kept the same in each experiment.

The table shows the results obtained in one series of experiments.

Concentration of hydrochloric acid / mol/dm ³	0.6	0.8	1.0	1.6	1.8	2.0
balance reading/ g	-0.20	-0.27	-0.44	-0.54	-0.60	-0.67

State why the balance readings have negative values.

.....

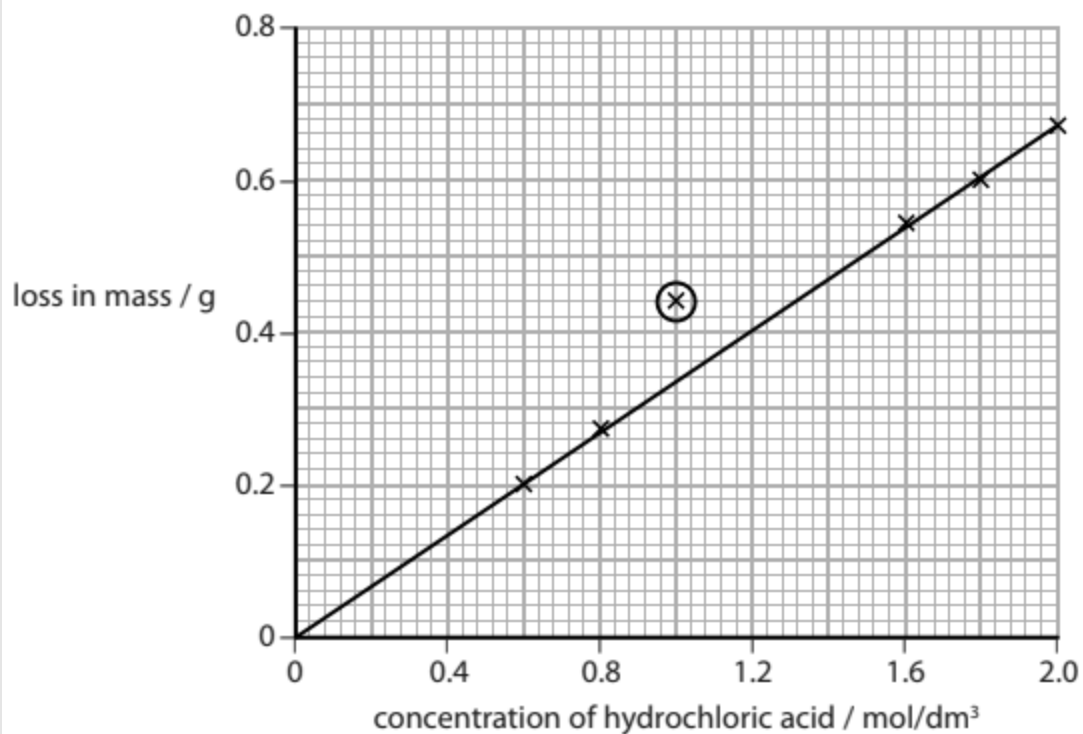
.....

.....

(b) multiple

[2]

The graph shows the results of this series of experiments.



The circled point indicates an anomalous result.

i) Suggest **one** mistake the students could have made to produce this result.

(1)

ii) State the relationship shown by the graph.

(1)

(c) explain

[2]

Explain why an increase in the concentration of the acid causes an increase in the rate of the reaction. You should use the particle collision theory in your answer.

Question 141 (5 marks)

(a) give

[2]

Welding involves joining two pieces of metal together.

Copper and iron are two metals that this can occur with.

Give two properties typical of these metals.

(b)

[1]

Metals have a lattice structure? What does this lattice consist of?

- A) anions and cations
- B) anions
- C) cations in a sea of delocalised electrons
- D) molecules in a sea of electrons

(c) state

[1]

To join two pieces of metal by welding, they must be melted together.

State why a high temperature has to be used.

(d)

[1]

An atmosphere of argon is used when the pieces of metal are welded together. Why is an atmosphere of argon used?

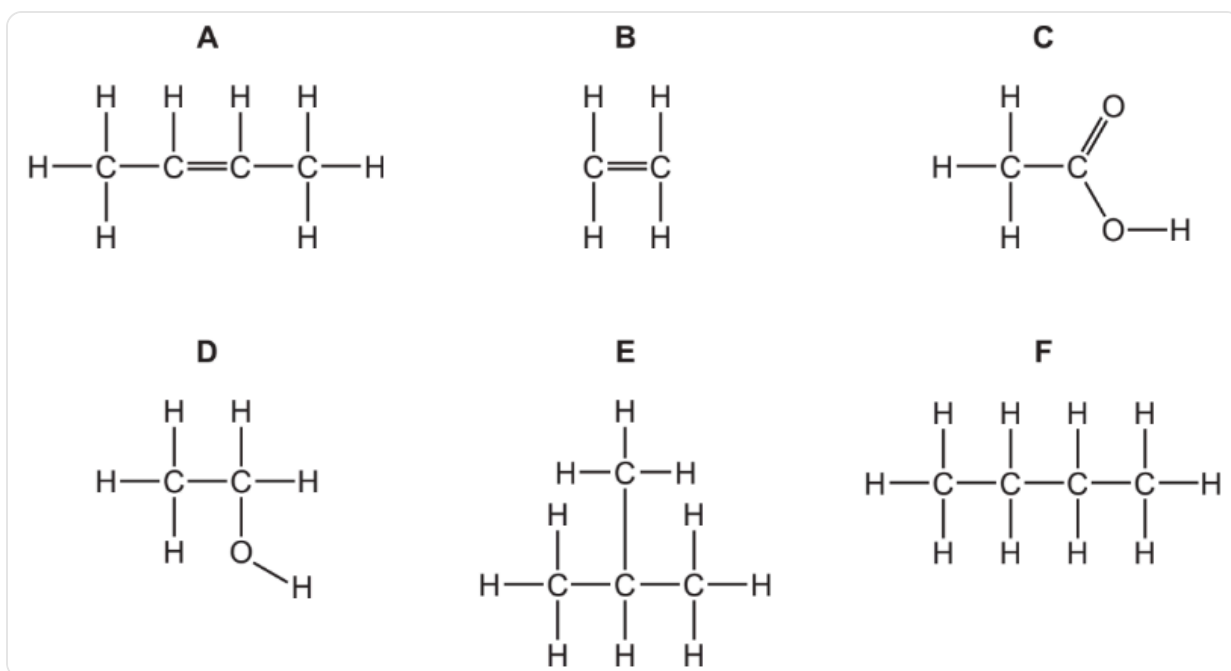
- A) to speed up the process of welding
- B) to prevent the metal reacting with oxygen
- C) to increase the melting point of the materials
- D) to save energy

Question 142 (6 marks)

(a) give

[1]

The structures of six organic compounds are shown.



Give the name of **F**.

.....

.....

(b) identify

[1]

Identify two compounds that are members of the same homologous series.

.....

.....

(c) state

[2]

Compounds **E** and **F** are isomers of each other.

State what is meant by the term **isomers**.

.....

.....

(d) explain

[2]

Explain why **B** is an unsaturated hydrocarbon.

.....

.....

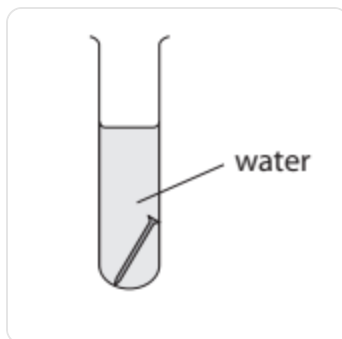
Question 143 (6 marks)

(a) multiple

[2]

A student investigates the rusting of iron.

She places an iron nail in a test tube of water and leaves it for several days.



i) Predict the appearance of the iron nail after several days.

(1)

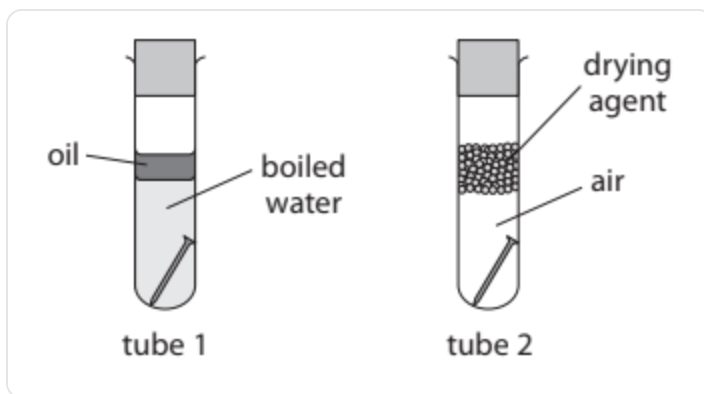
ii) Name the main compound in rust.

(1)

(b) explain

[4]

The student then sets up two more test tubes containing iron nails.



Explain why the iron nail in tube 1 and the iron nail in tube 2 do not rust.

tube 1

tube 2

Question 144 (6 marks)

(a) what

[1]

This question is about covalent bonding.

What is a covalent bond?

.....

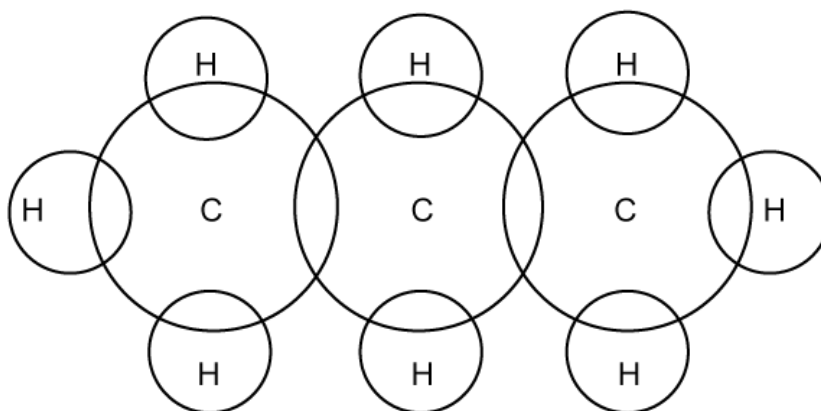
.....

.....

(b) complete

[1]

Complete the dot and cross diagram in to show the bonding in propane.



(1)

.....

.....

.....

(c) explain

[2]

Explain why propane cannot conduct electricity.

.....

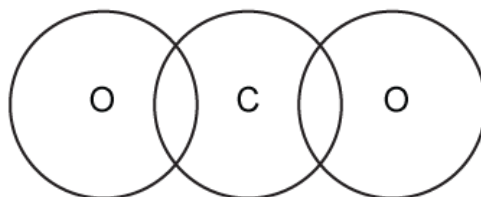
.....

.....

(d) complete

[2]

Carbon dioxide, CO_2 , bonds in a similar way to propane. Complete the diagram to show the bonding.



.....

.....

.....

Question 145 (6 marks)

(a) multiple

[2]

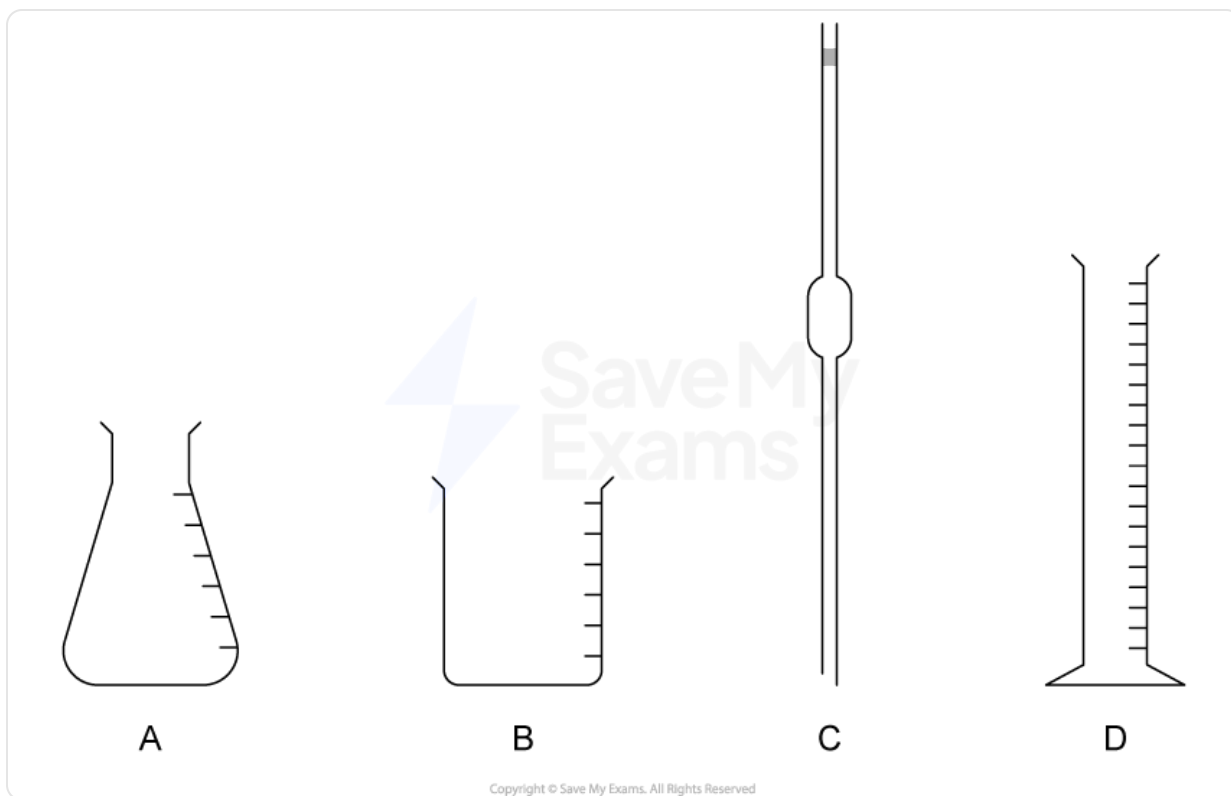
A student performs a titration to determine the concentration of an unknown solution of sodium hydroxide.

The student's method is

- 25.0 cm^3 of alkali is added to a clean conical flask
- A few drops of indicator are added, and the conical flask is placed on a white tile so that the colour change can be seen more easily
- The burette is filled with acid, and the starting volume is recorded
- The acid is slowly added from the burette to the alkali in the conical flask while the mixture is swirled
- The addition of acid is stopped when the end-point is reached (when the indicator changes colour), and the final volume is recorded
- The steps are repeated until concordant results are obtained

i) Which letter represents the apparatus the student should use to measure exactly 25.0 cm^3 of the sodium hydroxide solution?

[1]



ii) Which is the best indicator to use?

[1]

A. litmus

B. methyl orange

C. universal indicator

D. iodine solution

(b) multiple

[4]

The student obtains the following results:

Volume of sodium hydroxide solution used in cm^3	25.0
Concentration of sodium hydroxide solution in mol/dm^3	0.80
Mean volume of sulfuric acid solution used in cm^3	13.50

i) Write a balanced equation for the reaction between sulfuric acid, H_2SO_4 , and sodium hydroxide, NaOH .

[1]

ii) Calculate the amount, in moles, of sodium hydroxide in 25.0 cm^3 .

[1]

iii) Calculate the amount, in moles, of sulfuric acid in 13.50 cm^3 .

[1]

iv) Calculate the concentration of the sulfuric acid in mol/dm^3 .

[1]

Question 146 (6 marks)

(a) give [1]

A student does these two tests on a solution made from a white solid.

- flame test
- add acidified silver nitrate solution

The table shows his results.

Test	Result
flame test	red flame
add acidified silver nitrate solution	cream precipitate

Give the formula of the ion that produces the red flame.

(b) name [1]

Name the cream precipitate.

(c) identify [1]

Identify the white solid.

(d) multiple [3]

The student uses a clean metal wire in the flame test.

i) State why the wire should be clean when used in the flame test.

(1)

ii) The table lists properties of some metals.

Add ticks (✓) to the table to show the two properties needed in a metal wire used in a flame test.

(2)

Property
good conductor of electricity
high density
high melting point
unreactive

Question 147 (6 marks)

(a) complete

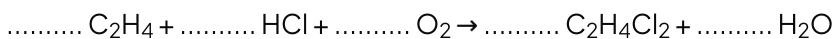
[1]

Ethene (C_2H_4) can be converted into chloroethene ($\text{C}_2\text{H}_3\text{Cl}$) in a two-stage process.

The first stage is to convert ethene into 1,2-dichloroethane, $\text{C}_2\text{H}_4\text{Cl}_2$

Ethene is reacted with hydrogen chloride and oxygen.

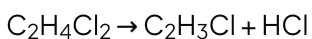
Complete the chemical equation for this reaction.



(b) state

[1]

In the second stage, 1,2-dichloroethane is converted into chloroethene.



This is a thermal decomposition reaction.

State what is meant by the term **thermal decomposition**.

.....

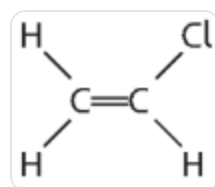
.....

.....

(c) multiple

[3]

The diagram shows the displayed formula of chloroethene.



i) State why chloroethene is described as an unsaturated compound.

(1)

ii) Describe a test to show that chloroethene is unsaturated.

(2)

.....

.....

.....

(d) name

[1]

Name the polymer formed from chloroethene.

.....

.....

.....

Question 148 (6 marks)

(a) how

[1]

This question is about reactions of iron.

Iron can form iron(III) oxide with a chemical formula of Fe_2O_3 .

How many atoms are shown in the formula Fe_2O_3 ?

(b) calculate

[1]

Calculate the relative formula mass iron(II) hydroxide, $\text{Fe}(\text{OH})_2$

(A_r : Fe = 56, O = 16, H = 1)

(c) calculate

[1]

Calculate the percentage by mass of iron in iron(II) hydroxide, $\text{Fe}(\text{OH})_2$.

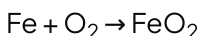
(d) draw

[3]

Recent discoveries in the fields of Geology and Seismology have established the potential existence of a previously undiscovered iron oxide compound, FeO_2 .

This iron oxide requires very high pressures and temperatures to form.

In theory, FeO_2 could be formed by the reaction of iron and oxygen, under suitable conditions, as follows.



How does this reaction demonstrate the law of conservation of mass?

Draw a ring around the correct answers to complete the sentences.

There is/are **one** ~~two~~ ~~three~~ atom(s) of iron on both sides of the equation

There is/are **one** ~~two~~ ~~three~~ atom(s) of oxygen on both sides of the equation

The total number of atoms **increases** ~~stays the same~~ ~~decreases~~

Question 149 (6 marks)

(a) name [1]

The box lists some substances. Choose substances from the box to answer these questions.

Each substance may be used once, more than once or not at all.

air	bromine	carbon	copper	glucose
	nitrogen	oxygen	sulfur	water

Name a metallic element.

.....

.....

(b) name [1]

Name a compound.

.....

.....

(c) name [1]

Name a mixture.

.....

.....

(d) name [1]

Name an element that is a gas at room temperature.

.....

.....

(e) name [1]

Name an element that forms a basic oxide.

.....

.....

(f) name [1]

Name two elements that are in the same group of the Periodic Table.

.....

.....

Question 150 (6 marks)

(a)

[1]

Methanol and ethanol are both members of the same homologous series.

Which of the following is the functional group of this homologous series?

- A) C=C
- B) -COOH
- C) -OH
- D) C-C

(b)

[1]

Which of the following is **not** a feature of a homologous series?

- A) the compounds differ by CH_2 in molecular formula from neighbouring compounds
- B) they have similar chemical properties
- C) they have the same general formula
- D) they have the same physical properties

(c) complete

[2]

Ethanoic acid can be produced from ethanol.

Use words from the box to complete the sentences about ethanol and ethanoic acid.

-OH	oxidised	-COOH	displaced
reduced	C=C	neutralised	decomposed

Ethanol can be to form ethanoic acid.

The functional group of ethanoic acid is

.....

.....

(d) draw

[2]

Like methanol and ethanol, butanol is an alcohol. Draw the structure of butanol. You should show all of the bonds in the structure.

.....

.....

.....

Question 151 (6 marks)

(a) give

[1]

The alcohols form a homologous series.

Give the name of the third member of this series.

.....

.....

.....

(b) calculate

[2]

Calculate the relative formula mass of hexanol, $C_6H_{13}OH$.

(A_r : C = 12, H = 1, O = 16)

relative formula mass =

.....

.....

.....

(c) place

[2]

Alcohols can be oxidised to form organic compounds from a different homologous series.

Place ticks (✓) in boxes by the two statements that are correct

Alkenes are formed when alcohols are oxidised
Ethanol oxidises to form ethanoic acid
Alcohols can be oxidised by heating with potassium dichromate(VI) in dilute sulfuric acid
Alcohols are oxidised by fermenting at 30 °C with the absence of oxygen
When ethanol oxidises in air, carbon dioxide is produced

.....

.....

.....

(d)

[1]

The structural formula of the first four alcohols in the homologous series are shown below, in order.

- CH_3OH
- C_2H_5OH
- C_3H_7OH
- C_4H_9OH

Which of the following structural formulae is the correct formula for the fifth member of the series?

- A) $C_{₅}H_{₁₀}OH$
- B) $C_{₅}H_{₁₁}COOH$
- C) $C_{₅}H_{₁₁}OH$
- D) $C_{₅}H_{₁₃}OH$

Question 152 (6 marks)

(a) multiple

[2]

Ethyl ethanoate has a distinctive smell which is similar to pear drops.

It is made by the reaction between ethanol and ethanoic acid in the presence of a sulfuric acid catalyst.

i) State the name that is given to this process.

(1)

ii) Which of the following is the other product formed during this reaction?

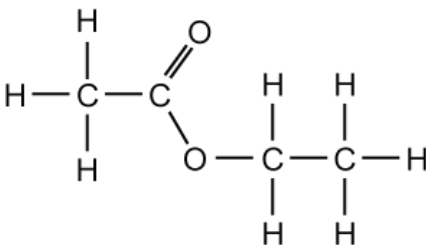
(1)

A	water
B	hydrogen chloride
C	hydroxide
D	hydrogen

(b) calculate

[2]

The displayed formula of ethyl ethanoate is shown below.



Calculate the relative molecular mass of ethyl ethanoate.

(A_r : C = 12, H = 1, O = 16)

(c) give

[1]

The reaction between ethanoic acid and ethanol to form ethyl ethanoate and another product is a reversible reaction.

Give the sign that is used in reactions to show that a reaction is reversible.

(d)

[1]

The reaction between ethanoic acid and ethanol will reach dynamic equilibrium.

Which statement about the dynamic equilibrium of this reaction is correct?

- A) the concentration of the reactants equals the concentration of the products
- B) equilibrium is established in an open container to allow the gases to escape
- C) the reaction has stopped
- D) the rate of the forward reaction is equal to the rate of the reverse reaction

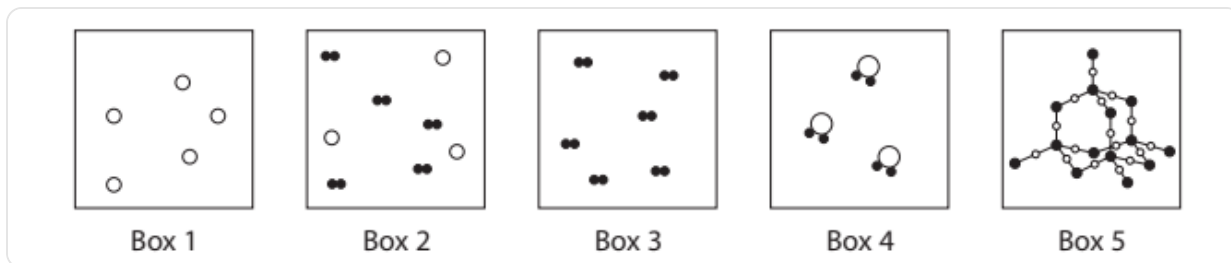
Question 153 (6 marks)

(a) multiple

[4]

Substances can be classified as elements, mixtures or compounds.

Each box represents an element, a mixture or a compound.



i) Which box represents a mixture?

☐	A	1
☐	B	2
☐	C	3
☐	D	4

(1)

ii) Which two boxes represent elements?

☐	A	1 and 2
☐	B	2 and 3
☐	C	1 and 3
☐	D	3 and 4

(1)

iii) Explain why Box 5 represents a compound.

(2)

.....

.....

.....

(b) multiple

[2]

The Periodic Table contains all the known elements.

i) How are the elements arranged in the Periodic Table?

☐	A	increasing mass number
☐	B	Increasing number of neutrons
☐	C	increasing number of protons
☐	D	increasing reactivity

(1)

ii) Elements in the same group have the same number of

☐	A	electrons in outer shell
☐	B	electron shells
☐	C	neutrons
☐	D	protons

(1)

Question 154 (6 marks)

(a)

[1]

The alkanes are a family of hydrocarbons with the general formula C_nH_{2n+2} . Ethane, C_2H_6 , and butane, C_4H_{10} , are examples of alkanes.

Ethane has a boiling point of -89°C and butane has a higher boiling point of -1°C .

Why is the boiling point of butane higher than ethane?

- A) Butane is ionically bonded
- B) Butane has a higher r than ethane
- C) Butane has stronger bonds between the carbon and hydrogen atoms
- D) Ethane has a higher r than butane

(b) complete

[3]

Complete the sentences below using words from the box.

conduct atoms free shells

fixed insulate ions electrons

Ethane and butane do not _____ electricity because there are no
_____ ions / _____ that are able to move and carry charge.

(c) what

[1]

What is the type of bonding in ethane and butane?

(d) multiple

[1]

Ethane and butane only contains single bonds. Double bonds can also form between a different family of hydrocarbons called alkenes. Which of the following statements is correct?

Tick () **one** box.

A double bond contains two electrons

A double bond contains two pairs of electrons